EUROPEAN COMMISSION
Executive Agency for Small and Medium-sized Enterprises
**Director**

**AMENDMENT Reference No AMD-776708-9**

**Grant Agreement number: 776708 — Innovative circular solutions and services for new business opportunities in the EU housing sector (HOUSEFUL)**

The parties agree to amend the Grant Agreement as follows ('**Amendment**'):

## 1 . Change of beneficiary due to partial takeover

As from 1 February 2019 the Grant Agreement is transferred from:

- **VISUM LIMITED (VISUM)**, established in COOLKERAGH LOWER, LISTOWEL KERRY, Ireland, VAT number: IE3230958KH, (the 'former beneficiary').

to

- **IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA (IRIS)**, established in CALLE VELAZQUEZ, NO 94 PRIMERA PLANTA, MADRID 28006, Spain, VAT number: ESB64446123, (the 'new beneficiary' with beneficiary number 17), if it acceded to the Agreement by signing the Accession Form (see Annex 3) in the electronic exchange system (see Article 52).

This implies the **following changes** to the Grant Agreement:

- The 'transfer date' and the new beneficiary are added to the **Preamble**:

  "17. **IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA (IRIS)**, established in CALLE VELAZQUEZ, NO 94 PRIMERA PLANTA, MADRID 28006, Spain, VAT number: ESB64446123, — as from 1 February 2019."

  "6. **VISUM LIMITED (VISUM)**, established in COOLKERAGH LOWER, LISTOWEL KERRY, Ireland, VAT number: IE3230958KH, — until 31 January 2019."

The former beneficiary's obligations continue to apply after the transfer (in particular, Articles 20, 22, 23, Section 3 of Chapter 4, 36, 37, 38, 40, 42, 43 and 44), although it no longer participates in the action.

The new beneficiary(ies) take over all the rights and obligations of the former beneficiary under the Grant Agreement and have full responsibility for implementing the action and complying with the Agreement. In addition, the new beneficiary is jointly and severally liable for any debts of the former beneficiary, concerning recoveries.





The joint and several liability of the linked third parties of the former beneficiary(ies) continues to apply after the transfer, even if they no longer participate in the action.

**2. Change of Annex 1 (description of the action)**

**Annex 1** is changed and replaced by the Annex 1 attached to this Amendment.

**3 . Changes of Annex 2 (estimated budget)**

**Annex 2** is changed and replaced by the Annex 2 attached to this Amendment.

All other provisions of the Grant Agreement and its Annexes remain unchanged.

This Amendment **enters into force** on the day of the last signature.

This Amendment **takes effect** on the date on which the amendment enters into force, except where a different date has been agreed by the parties (for one or more changes).

Please inform the other members of the consortium of the Amendment.

SIGNATURES

For the coordinator | For the Agency

Enclosures:

Annex 2
Annex 1

**ANNEX 1 (part A)**

**Innovation action**

**NUMBER — 776708 — HOUSEFUL**

# Table of Contents

1.1. The project summary....3
1.2. The list of beneficiaries....4
1.3. Workplan Tables - Detailed implementation....5
1.3.1. WT1 List of work packages....5
1.3.2. WT2 List of deliverables....6
1.3.3. WT3 Work package descriptions....12
Work package 1.... 12
Work package 2.... 16
Work package 3.... 21
Work package 4.... 25

Work package 5 ..... 36
Work package 6 ..... 40
Work package 7 ..... 47
Work package 8 ..... 54
1.3.4. WT4 List of milestones ..... 56
1.3.5. WT5 Critical Implementation risks and mitigation actions ..... 58
1.3.6 WT6 Summary of project effort in person-months ..... 65
1.3.7. WT7 Tentative schedule of project reviews ..... 66

# 1.1. The project summary

| Project Number [1] | 776708 | Project Acronym [2] | HOUSEFUL |
|---|---|---|---|

One form per project

General information

| | |
|---|---|
| Project title [3] | Innovative circular solutions and services for new business opportunities in the EU housing sector |
| Starting date [4] | 01/05/2018 |
| Duration in months [5] | 54 |
| Call (part) identifier [6] | H2020-CIRC-2017TwoStage |
| Topic | CIRC-01-2016-2017<br>Systemic, eco-innovative approaches for the circular economy: large-scale demonstration projects |
| Fixed EC Keywords | Resources efficiency, Circular economy, Service innovation, Technological innovation, Energy Efficiency, Business model innovation |
| Free keywords | Social innovation, systemic service, housing sector, construction, water re-use, biowaste treatment, biogas production, wastewater, co-creation , circular economy business opportunities |

Abstract [7]

The housing sector is a major contributor to current global problems of resource depletion and climate change, representing one of the most important consuming sectors at EU level: 50% of all extracted materials, 40% of final energy consumption, 33% of water consumption and 33% of all produced waste. The lock-in to the linear business models of today is causing many environmental problems and is one of the major barriers in transition towards a circular economy. HOUSEFUL project proposes an innovative paradigm shift towards a circular economy for the housing sector by demonstrating the feasibility of an integrated systemic service composed of 11 circular solutions. HOUSEFUL will introduce solutions to become more resource efficient throughout the lifecycle of a building, taking into account an integrated circular approach where energy, materials, waste and water aspects are considered. This approach fosters new forms of co-creation, increasing the collaboration among stakeholders of the housing value chain to develop new circular solutions and services. HOUSEFUL concept will be large scale demonstrated at 4 demo-sites in Austria and Spain, adapting the concept to different scenarios, including in social housing buildings. HOUSEFUL solutions will be evaluated from an environmental (Life Cycle Assessment), economic (Life Cycle Cost) and social (Social Assessment) point of view. The results obtained will be used to define an integrated HOUSEFUL service which will be driven and promoted through a SaaS (Software as a Service). The SaaS will integrate a
Circularity Tool to quantify the circularity level of buildings and will include different circular solutions to be offered as services, encouraging the housing value chain to redesign traditional business models towards circular ones. 10 EU Follower buildings will be engaged with the support of a Collaborative Community of Housing Experts to replicate HOUSEFUL results and maximise the impact of the project.

# 1.2. List of Beneficiaries

| Project Number [1] | 776708 | Project Acronym [2] | HOUSEFUL |
|---|---|---|---|

List of Beneficiaries

| No | Name | Short name | Country | Project entry date[8] | Project exit date |
|---|---|---|---|---|---|
| 1 | ACONDICIONAMIENTO TARRASENSE ASSOCIACION | LEITAT | Spain | | |
| 2 | INSTITUT DE TECNOLOGIA DE LA CONSTRUCCION DE CATALUNYA | ITeC | Spain | | |
| 3 | FUNDACION CARTIF | CARTIF | Spain | | |
| 4 | ALCHEMIA-NOVA GMBH | ALCN | Austria | | |
| 5 | AGENCIA DE L'HABITATGE DE CATALUNYA | AHC | Spain | | |
| 6 | VISUM LIMITED | VISUM | Ireland | | 31/01/2019 |
| 7 | SISTEMES AVANCATS DE ENERGIA SOLAR TERMICA SCCL AIGUASOL | AIGUASOL | Spain | | |
| 8 | LGI CONSULTING | LGI | France | | |
| 9 | ARCHITEKTURBURO REINBERG ZT GMBH | GWR | Austria | | |
| 10 | TURNTOO B.V. | Turntoo B.V. | Netherlands | | |
| 11 | GEMEINNUTZIGE BAU-, WOHN UND SIEDLUNGSGENOSSENSCHAFT NEUES LEBEN REGISTRIERTE GENOSSENCHAFT MIT BESCHRANKTER HAFTUNG | Neues Leben | Austria | | |
| 12 | COMITE EUROPEEN DE COORDINATION DE L'HABITAT SOCIAL AISBL | HOUSING EUROPE | Belgium | | |
| 13 | WATER, ENVIRONMENT AND BUSINESS FORDEVELOPMENT SL | WE&B | Spain | | |
| 14 | FONDAZIONE ICONS | ICONS | Italy | | |
| 15 | IDP INGENIERIA Y ARQUITECTURA IBERIA SL | IDP | Spain | | |
| 16 | HOMEBIOGAS LTD | HOMEBIOGAS LTD | Israel | | |
| 17 | IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA | IRIS | Spain | 01/02/2019 | |

# 1.3. Workplan Tables - Detailed implementation

## *1.3.1. WT1 List of work packages*

| WP Number[9] | WP Title | Lead beneficiary[10] | Personmonths[11] | Start month[12] | End month[13] |
|---|---|---|---|---|---|
| WP1 | Project Management | 1 - LEITAT | | 1 | 54 |
| WP2 | Methodology for the evaluation of circularity in the housing sector | 7 - AIGUASOL | | 1 | 48 |
| WP3 | Social engagement: starting the cocreation of HOUSEFUL services | 13 - WE&B | | 1 | 54 |
| WP4 | Large scale demo buildings | 3 - CARTIF | | 3 | 48 |
| WP5 | Environmental, economic and social assessment | 1 - LEITAT | | 8 | 54 |
| WP6 | HOUSEFUL Solutions as Services, market, policies and exploitation | 8 - LGI | | 1 | 54 |
| WP7 | Communication and Dissemination | 14 - ICONS | | 1 | 54 |
| WP8 | Ethics requirements | 1 - LEITAT | | 1 | 54 |
| | | **Total** | | | |

### *1.3.2. WT2 list of deliverables*

| Deliverable Number[14] | Deliverable Title | WP number[9] | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|---|
| D1.1 | Project Manual | WP1 | 1 - LEITAT | Report | Confidential, only for members of the consortium (including the Commission Services) | 1 |
| D1.2 | Data Management Plan (DMP) | WP1 | 1 - LEITAT | ORDP: Open Research Data Pilot | Confidential, only for members of the consortium (including the Commission Services) | 6 |
| D1.3 | First activity and management report | WP1 | 1 - LEITAT | Report | Confidential, only for members of the consortium (including the Commission Services) | 12 |
| D1.4 | Final DMP | WP1 | 1 - LEITAT | ORDP: Open Research Data Pilot | Confidential, only for members of the consortium (including the Commission Services) | 54 |
| D1.5 | Minutes of management meetings | WP1 | 1 - LEITAT | Report | Confidential, only for members of the consortium (including the Commission Services) | 54 |
| D2.1 | Report on end-users requirements and CEBO on residential buildings | WP2 | 7 - AIGUASOL | Report | Public | 6 |
| D2.2 | Circular economy repository for building archetypes in the geoclusters | WP2 | 7 - AIGUASOL | Report | Confidential, only for members of the consortium (including the Commission Services) | 12 |

| | | | | | | |
|---|---|---|---|---|---|---|
| D2.3 | Reference methodologies and KPIs in circular economy analysis | WP2 | 7 - AIGUASOL | Report | Public | 18 |
| D2.4 | Existing market circularity solutions and product | WP2 | 7 - AIGUASOL | Report | Public | 26 |

| **Deliverable Number[14]** | **Deliverable Title** | **WP number[9]** | **Lead beneficiary** | **Type[15]** | **Dissemination level[16]** | **Due Date (in months)[17]** |
|---|---|---|---|---|---|---|
| D2.5 | First approach of BCM methodology and implementation in theoretical scenarios | WP2 | 7 - AIGUASOL | Report | Public | 26 |
| D2.6 | Description of the BCM methodology and implementation rules in real scenarios | WP2 | 7 - AIGUASOL | Report | Public | 48 |
| D3.1 | Social engagement strategy for the co-creation of HOUSEFUL solutions as new services (version I) | WP3 | 13 - WE&B | Report | Public | 18 |
| D3.2 | Social perceptions and beliefs in circular economy at building scale (version I) | WP3 | 13 - WE&B | Report | Public | 32 |
| D3.3 | Social conditions for the co-creation of all HOUSEFUL services and co-created materials | WP3 | 13 - WE&B | Report | Public | 48 |
| D3.4 | Social engagement strategy for the co-creation of HOUSEFUL solutions as new services in Demo 2 and Demo 3 | WP3 | 13 - WE&B | Report | Public | 26 |
| D3.5 | Social perceptions and beliefs in circular economy at building scale (version II - Final) | WP3 | 13 - WE&B | Report | Public | 42 |

| | | | | | | |
|---|---|---|---|---|---|---|
| D3.6 | Social engagement strategy for the co-creation of HOUSEFUL solutions as new services (Final) | WP3 | 13 - WE&B | Report | Public | 42 |
| D4.1 | Circular economy audit: Baseline calculation for Demo 1 and 4 | WP4 | 3 - CARTIF | Report | Public | 18 |
| D4.2 | Technical specifications and execution project of solutions in Spanish and Austrian demos: Demo 1 and Demo 4 | WP4 | 3 - CARTIF | Demonstrator | Confidential, only for members of the consortium (including the Commission Services) | 18 |

| **Deliverable Number[14]** | **Deliverable Title** | **WP number[9]** | **Lead beneficiary** | **Type[15]** | **Dissemination level[16]** | **Due Date (in months)[17]** |
|---|---|---|---|---|---|---|
| D4.3 | BIM models and material passport of the Spanish and Austrian demos: Demo 1 and Demo 4 | WP4 | 2 - ITeC | Report | Public | 18 |
| D4.4 | Validation tests of technologies in S5, S6, S7 and S8 solutions at lab scale. Includes the Safe-ty and monitoring plan for S7 and S8 | WP4 | 1 - LEITAT | Report | Public | 36 |
| D4.5 | Report on potential improvement in financial schemes and funding opportunities for Demo 2 Frontrunner buildings and replication activities | WP4 | 12 - HOUSING EUROPE | Report | Confidential, only for members of the consortium (including the Commission Services) | 26 |
| D4.6 | Report on potential improvement in financial schemes and funding opportunities for Demo 3 Frontrunner buildings and replication activities | WP4 | 12 - HOUSING EUROPE | Report | Confidential, only for members of the consortium (including the Commission Services) | 30 |
| D4.7 | Report of demonstration and implementation of HOUSEFUL solutions at building scale. Includes the Technical monitoring programme | WP4 | 3 - CARTIF | Demonstrator | Public | 48 |

| | | | | | | |
|---|---|---|---|---|---|---|
| D4.8 | Report on financial schemes and funding opportunities for all Front-runner buildings and replication activities (final update) | WP4 | 12 - HOUSING EUROPE | Report | Confidential, only for members of the consortium (including the Commission Services) | 48 |
| D4.9 | Technical specifications and execution project of solutions in Spanish and Austrian demos (final version) | WP4 | 3 - CARTIF | Demonstrator | Confidential, only for members of the consortium (including the Commission Services) | 36 |
| D4.10 | BIM models and Material passport for the Spanish and Austrian demos integrating HOUSEFUL solutions | WP4 | 2 - ITeC | Report | Public | 48 |

| **Deliverable Number[14]** | **Deliverable Title** | **WP number[9]** | **Lead beneficiary** | **Type[15]** | **Dissemination level[16]** | **Due Date (in months)[17]** |
|---|---|---|---|---|---|---|
| D4.11 | Circular economy audit: Baseline calculation for Demo 2 and Demo 3 | WP4 | 3 - CARTIF | Report | Public | 23 |
| D4.12 | Technical specifications and execution project of solutions in Spanish and Austrian demos: Demo 2 and Demo 3 | WP4 | 3 - CARTIF | Demonstrator | Confidential, only for members of the consortium (including the Commission Services) | 26 |
| D4.13 | BIM models and material passport of the Spanish and Austrian demos: Demo 2 and Demo 3 | WP4 | 2 - ITeC | Report | Confidential, only for members of the consortium (including the Commission Services) | 30 |
| D5.1 | HOUSEFUL assessment plan | WP5 | 3 - CARTIF | Report | Public | 17 |
| D5.2 | Environmental and economic impact assess-ment | WP5 | 1 - LEITAT | Report | Public | 54 |
| D5.3 | Measurement &Verification | WP5 | 1 - LEITAT | Report | Public | 54 |
| D5.4 | Social assessment | WP5 | 13 - WE&B | Report | Public | 54 |

| | | | | | | |
|---|---|---|---|---|---|---|
| D6.1 | Market study (version I) | WP6 | 8 - LGI | Report | Public | 18 |
| D6.2 | Solution-to-Service monitoring plan (version I) | WP6 | 1 - LEITAT | Report | Confidential, only for members of the consortium (including the Commission Services) | 18 |
| D6.3 | HOUSEFUL SaaS | WP6 | 17 - IRIS | Other | Confidential, only for members of the consortium (including the Commission Services) | 36 |
| D6.4 | HOUSEFUL SaaS – Final version | WP6 | 17 - IRIS | Other | Public | 54 |
| D6.5 | Impact study of HOUSEFUL services on linear business models | WP6 | 10 - Turntoo B.V. | Report | Public | 36 |
| D6.6 | Preliminary Exploitation Plan | WP6 | 8 - LGI | Report | Confidential, only for members of the consortium (including the | 36 |

| Deliverable Number[14] | Deliverable Title | WP number[9] | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|---|
| | | | | | Commission Services) | |
| D6.7 | Replicability analysis of new CEBOs | WP6 | 2 - ITeC | Report | Confidential, only for members of the consortium (including the Commission Services) | 48 |
| D6.8 | Final Exploitation Plan | WP6 | 8 - LGI | Report | Confidential, only for members of the consortium (including the Commission Services) | 54 |
| D6.9 | Policy recommendations | WP6 | 12 - HOUSING EUROPE | Report | Public | 54 |
| D6.10 | Market study (version II - final) | WP6 | 8 - LGI | Report | Public | 24 |

| | | | | | | |
|---|---|---|---|---|---|---|
| D6.11 | Solution-to-Service monitoring plan (version II) | WP6 | 1 - LEITAT | Report | Confidential, only for members of the consortium (including the Commission Services) | 36 |
| D6.12 | Solution-to-Service monitoring plan (version III - final) | WP6 | 1 - LEITAT | Report | Confidential, only for members of the consortium (including the Commission Services) | 51 |
| D7.1 | Dissemination & Communication plan | WP7 | 14 - ICONS | Report | Confidential, only for members of the consortium (including the Commission Services) | 43 |
| D7.2 | HOUSEFUL website | WP7 | 14 - ICONS | Websites, patents filling, etc. | Public | 5 |
| D7.3 | Project presentation video | WP7 | 14 - ICONS | Websites, patents filling, etc. | Public | 8 |
| D7.4 | Call-To-Action video | WP7 | 14 - ICONS | Websites, patents filling, etc. | Public | 18 |
| D7.5 | Flyers (Communication materials for stakeholder engagement) | WP7 | 14 - ICONS | Report | Public | 8 |

| **Deliverable Number[14]** | **Deliverable Title** | **WP number[9]** | **Lead beneficiary** | **Type[15]** | **Dissemination level[16]** | **Due Date (in months)[17]** |
|---|---|---|---|---|---|---|
| D7.6 | Project leaflet | WP7 | 14 - ICONS | Report | Public | 10 |
| D7.7 | Report on public communication, networking and clustering activities (version I) | WP7 | 14 - ICONS | Report | Public | 18 |
| D7.8 | Multimedia tools: 1 PageFlow, 1 VNR and 1 web video produced and distributed | WP7 | 14 - ICONS | Websites, patents filling, etc. | Public | 50 |
| D7.9 | Best Practices book | WP7 | 14 - ICONS | Websites, patents filling, etc. | Public | 52 |

| | | | | | | |
|---|---|---|---|---|---|---|
| D7.10 | Communication and dissemination activities impact assessment | WP7 | 14 - ICONS | Report | Confidential, only for members of the consortium (including the Commission Services) | 54 |
| D7.11 | Report on public communication, networking and clustering activities (version II) | WP7 | 14 - ICONS | Report | Public | 36 |
| D7.12 | Report on public communication, networking and clustering activities (version III - final) | WP7 | 14 - ICONS | Report | Public | 54 |
| D8.1 | H - Requirement No. 1 | WP8 | 1 - LEITAT | Ethics | Confidential, only for members of the consortium (including the Commission Services) | 1 |
| D8.2 | POPD - H - Requirement No. 3 | WP8 | 1 - LEITAT | Ethics | Confidential, only for members of the consortium (including the Commission Services) | 1 |
| D8.3 | GEN - Requirement No. 4 | WP8 | 1 - LEITAT | Ethics | Confidential, only for members of the consortium (including the Commission Services) | 3 |

### *1.3.3. WT3 Work package descriptions*

| **Work package number** [9] | WP1 | **Lead beneficiary** [10] | 1 - LEITAT |
|---|---|---|---|
| **Work package title** | Project Management | | |
| **Start month** | 1 | **End month** | 54 |

**Objectives**

1) To coordinate and manage the project and the efficient, smooth and timely execution of all planned pro-ject activities.
2) To revise and maintain project work plan
3) To ensure an accurate and on time communication flow among the project partners and with EC in order to trackthe project progress and meet its objectives
4) To complete the project within the agreed time schedule and budget.

**Description of work and role of partners**

**WP1 - Project Management** [Months: 1-54]
**LEITAT**, ITeC, CARTIF, ALCN, AHC, VISUM, AIGUASOL, LGI, GWR, Turntoo B.V., Neues Leben, HOUSING EUROPE, WE&B, ICONS, IDP, HOMEBIOGAS LTD, IRIS
This work package deals with the coordination and management of all project activities and communication between partners and with the EC. The project will be coordinated by LEITAT (Project PC), with the help of a Management Committee, with a representative from each consortium partner, which will lead the management of the project, having the authority to make decisions on confidentiality and dissemination activities.

Task 1.1. Project Communication Flow (Leader: LEITAT; Participants: all) [M1-M54]
The PC will be in a continual contact with all the members of the Consortium and with the Work Packages leaders. The Management Committee meetings, defined in Management Section 3.2, will be also a communication context where administrative and financial assessment will be performed. In addition, the private area (intranet) of the project website described in the WP7 will be used for sharing management and financial related information, working documents, data, periodic reports and project deliverables which may be internally shared. Access to the project intranet will be limited to partners and the European Commission. The PC will also analyse potential and real conflicts between partners and propose agreements and make all necessary efforts to ensure a good and fluid communication process between all partners.

Task 1.2. Project Decision Structure Management (Leader: LEITAT; Participants: all) [M1-M54]
The decision structure is organised around the work package leadership. Each work package leader will be in charge of the specific tasks and deliverables for each work package. Each work package will be under the direct responsibility of the work package leader who will also draw guidelines for the work package-specific deliverables. This is expected to greatly assist the PC in the overall coordination of the project. The main decisions of the project will be taken by the Scientific Committee, which will be formed by representatives of each work package. These decisions will be implemented through regular meetings and correspondence among the consortium members, with 6 monthly updates sent to the consortium members through the PC. At each of the meetings, the consortium will discuss results of accomplished tasks, jointly develop a strategy for the next research step, and exchange general information regarding scientific work and its management. The day by day scientific progress of the project will be watched over by the Scientific & Technical Manager, who will as well review scientific reports and deliverables before being uploaded in the project's intranet. Any progress or alteration of the foreseen schedule will be reported to the Project PC, through a continuous and fluid communication, and to the Steering Committee, through half-yearly meetings to evaluate the Project's scientific development and the eventual correcting measures.

Task 1.3. Administrative and Financial Management (Leader: LEITAT; Participants: all) [M1-M54]
The PC will be responsible of the administrative coordination of the project. In the framework of this task, PC will monitor and steer the project implementation, paying special attention to the development of effective flows of communication within the Consortium and the efficient coordination of work between partners. The PC will collect all periodic activities and evaluation reports from all partners to be submitted and will inform all partners about suggestions and/or recommendations given by the Commission. The PC will also be responsible of the relations with the European Commission and of the correct distribution of the EC contribution between the partners. The PC will have to keep the Consortium informed of any information communicated by the European Commission. The PC will be responsible of the financial coordination of the project. This task will include the follow-up and delivery of all financial documents required periodically by the European Commission on time. The PC will also follow up the project expenses and track possible deviations. On the other hand, the PC will distribute the EC contribution amongst the partners according to the Grant Agreement and the Consortium Agreement.

Task 1.4. Data management plan (Leader: LEITAT; Participants: all) [M1-M54]
WP Leader will create a "Data Management Plan" (DMP) template using guidance provided by the EU. The aim of the DMP is to describe the data, generated and/or collected during the project life cycle and how it will be managed. Data refers to any kind of scientific result including statistics, results of experiments, measurements, observations resulting from fieldwork, survey results, interview recordings, images and bibliographic metadata used for the validation publications. According to EC recommendation, as part of making research data findable, accessible, interoperable and re-usable (FAIR), the DMP will include information on (1) the handling of research data during & after the end of the Project; (2) what data will be collected, processed and/or generated; (3) which methodology & standards will be applied; (4) whether data will be shared/made open Access; and (5) how data will be curated & preserved (including after the end of the project) in accordance with article 29.3 from the GA Option 1 for actions participating in the open Research Data Pilot, without changing the obligation to protect results, confidentiality, security and protection to personal data (Articles 27, 36, 37 and 39). HOUSEFUL project will always obtain informed consent from persons providing data which participation will be entirely voluntary. All activities procedures will be designed according to Directive 95/46/ EC of the European Parliament and of the Council of 24 October 1995 on the protection of individuals with regard to the processing of personal data and on the free movement of such data. Within the context of the HOUSEFUL project,

procedures for personal data must be processed in accordance with national and EU legislation, limiting the impact of the persons involved whilst ensuring data quality and confidentiality. Moreover, the DMP will include detailed information on the procedures for data collection, storage, protection, retention, and destruction.
WP Leader will draft a first version of the DMP, which will be delivered within the first 6 months and will be in charge to update it as data sets are produced during the project life. In order to improve and maximise open access and re-use of research data the PC, will update the DMP at a regular basis. It will take into ac-count the need to balance openness and protection of scientific information, commercialisation and Intellectual Property Rights (IPR), privacy concerns, security as well as data management. The implementation of the DMP will be coordinated and monitored by LEITAT, and the implementation will be done in coordination with LGI, since it is the partner in charge of the IPR management. The DMP will not have an impact on IPR, because the decision or whether to publish Open Access documents come after the more general decision on whether to first seek protection for intellectual property rights. Accordingly, research results can be published only after a patent application has been filled.

| Participation per Partner | |
|---|---|
| **Partner number and short name** | **WP1 effort** |
| 1 - LEITAT | |
| 2 - ITeC | |
| 3 - CARTIF | |
| 4 - ALCN | |
| 5 - AHC | |
| 6 - VISUM | |
| 7 - AIGUASOL | |
| 8 - LGI | |
| 9 - GWR | |
| 10 - Turntoo B.V. | |
| 11 - Neues Leben | |
| 12 - HOUSING EUROPE | |
| 13 - WE&B | |
| **Partner number and short name** | **WP1 effort** |
| 14 - ICONS | |
| 15 - IDP | |
| 16 - HOMEBIOGAS LTD | |
| 17 - IRIS | |
| **Total** | |

List of deliverables

| Deliverable Number[14] | Deliverable Title | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|
| D1.1 | Project Manual | 1 - LEITAT | Report | Confidential, only for members of the consortium (including the Commission Services) | 1 |
| D1.2 | Data Management Plan (DMP) | 1 - LEITAT | ORDP: Open Research Data Pilot | Confidential, only for members of the consortium (including the Commission Services) | 6 |
| D1.3 | First activity and management report | 1 - LEITAT | Report | Confidential, only for members of the consortium (including the Commission Services) | 12 |
| D1.4 | Final DMP | 1 - LEITAT | ORDP: Open Research Data Pilot | Confidential, only for members of the consortium (including the Commission Services) | 54 |
| D1.5 | Minutes of management meetings | 1 - LEITAT | Report | Confidential, only for members of the consortium (including the Commission Services) | 54 |

Description of deliverables

Deliverables: D1.1) Project Manual [M1, LEITAT] / D1.2) Data Management Plan (DMP) [M6, LEITAT] / D1.3) First activity and management report [M12, LEITAT]. / D1.4) Final DMP [M54, LEITAT] D1.5) Minutes of management meetings [M54, LEITAT]

D1.1 : Project Manual [1]

The Project Manual aims to aid project partners on their tasks and how to fulfil them. It is a dynamic reference document that will be updated as frequently as necessary. It provides details about: project organization, management structure, deliverables submission procedure, reports, intellectual property, publications and communication information. This deliverable is related to Task 1.1.

D1.2 : Data Management Plan (DMP) [6]

DMP aims at ensuring that the digital research data generated in the action will be made findable, accessible, interoperable and reusable in accordance with article 29.3 from the GA Option 1 for actions participating in the open Research Data Pilot, without changing the obligation to protect results, confidentiality, security and protection to personal data (Articles 27, 36, 37 and 39). IPR will be taken into account to ensure that making data accessible does not go against the interests of any beneficiary with regards of protection of intellectual property. HOUSEFUL project will always obtain informed consent from persons providing data which participation will be entirely voluntary. All activities procedures will be designed according to Directive 95/46/EC of the European Parliament and of the Council of 24 October 1995 on the protection of individuals with regard to the processing of personal data and on the free movement of such data. Within the context of the HOUSEFUL project, procedures for personal data must be processed in accordance with national and EU legislation, limiting the impact of the persons involved whilst ensuring data quality and confidentiality. This deliverable is related to Task 1.4.

D1.3 : First activity and management report [12]

A summary of the overall project status and specific state of progress of each WP will be prepared for each year of the project. The report will include possible deviations encountered and identified actions to achieve the project objectives. The Deliverable 1.3 will also provide an overview of the resources committed and explanations of major deviations in relation to the DoA. This deliverable is related to Task 1.3.

D1.4 : Final DMP [54]

Summary of the actions taken according to the first version of the DMP provided in D1.2 during the project execution period, evaluating its adequacy and the degree of fulfillment of foreseen objectives and procedures. Deliverable 1.4 will serve as an update of D1.2 and also will report on the degree of fulfillment of DMP within project execution. This deliverable is related to Task 1.4.

D1.5 : Minutes of management meetings [54]

Minutes of consortium meetings will be made accessible to all partners outlining the decisions taken at the meeting and the agreed plants for the next action of the project. Minutes will contain the signed list of attendees and will be submitted as D1.5 after each consortium meeting. This deliverable is related to Task 1.5.

## Schedule of relevant Milestones

| Milestone number[18] | Milestone title | Lead beneficiary | Due Date (in months) | Means of verification |
|---|---|---|---|---|

| | | | |
|---|---|---|---|
| **Work package number [9]** | WP2 | **Lead beneficiary [10]** | 7 - AIGUASOL |
| **Work package title** | Methodology for the evaluation of circularity in the housing sector | | |
| **Start month** | 1 | **End month** | 48 |

## Objectives

The main objective of WP2 is to develop a robust, reliable and day-to-day market usable methodology to quantify the degree of circularity of buildings, both pre- and post-refurbishment, which could be used by designers but also construction companies, promoters and financial institutions. The methodology will be developed considering the main circular economy principles related to recyclability, reusability and waste savings of the materials and active solutions defining the buildings, and also other key factors such as the feasibility of circular solutions (WP5) offered to stakeholders as services (WP6). The methodology will consider all the existing or brand-new methodologies related to the key vectors of circularity, including LCC, LCA, S-LCA methodologies or the next future brand-new EU Level(s) one. The methodology will result in an easy-to-use tool as HOUSEFUL Solution: S1, which will be the basis of SaaS developed in WP6.

Specific Objectives:

- To define the main weighting factors and KPIs representatives of the recyclability and reusability of materials, of theenergy and water life cycle, and of the social and economic performance;
- To define a set of rules and procedures from the existing bibliography and methodologies, and combine them in acomplete and comprehensive circularity calculation methodology;
- To establish some recommended and predefined values, both for representative parameters and also for required inputs,from residential building archetypes representing different European locations;
- To refine the methodology for the evaluation of circularity using the results obtained in the real demo scenarios andthe other WPs, establishing a high degree of reliability.

## Description of work and role of partners

**WP2 - Methodology for the evaluation of circularity in the housing sector** [Months: 1-48]

**AIGUASOL**, LEITAT, ITeC, CARTIF, ALCN, AHC, VISUM, GWR, Turntoo B.V., HOMEBIOGAS LTD, IRIS

Task 2.1. End users requirements considering circular economy business opportunities (Leader: LEITAT; Participants: AIGUASOL, ITeC, AHC, VISUM, ALCN, GWR) [M1-M6]

Task 2.1 aims to define the needs and requirements that the methodology for evaluating the circularity of buildings has to meet in order to provide the maximum value to the end-users. A valuable-by-design service will be therefore developed following a structured approached for defining customer needs and translating them into specific designs, according with the Quality Function Deployment (QFD) methodology. The "voice of the customer" will be collected via different channels such as: direct discussion or interviews, surveys, focus groups, customer specifications, observation, warranty data, field reports, etc. The output will be then elaborated in a development planning matrix called "house of quality". This matrix will be used to convert high-level needs into lower level features designed to satisfy these needs. VISUM will lead the implementation of the QFD methodology, providing templates, questionnaires and building the house of quality, while the other partners involved in the task will get in contact with the potential end-user and stakeholder in order to gather the information needed. AHC will provide input and feedback from the end user perspective with the aim of providing a useful tool for management of their property and housing programs. ALCN will support the survey of end user requirements concerning the demo building "Donaufelderstrasse" by organising communication with property developers (NEUES LEBEN) and dwellers. GWR will give input on building requirements and user acceptance and demand from everyday business. AIGUASOL will provide from end user requirements from other related projects (see Plug&Harvest or 4RinEU) and will adapt them to the HOUSEFUL frame, considering the expected new developments and SaaS services.

Task 2.2. Establishment of residential building archetypes for the representative European geoclusters (Leader: AIGUASOL; Participants: CARTIF) [M2-M12]

The aim of Task 2.2 is to establish the main circular economy conditions of building archetypes for the representative European geoclusters. The so-called circular economy main conditions will include the establishment of the main environmental aspects for each case (energy, water, materials and waste) that will be cross-referenced with the respective geoclusters, to obtain a set of representative circularity scenarios to be used in the following tasks of WP2, but also considered as a reference for other purposes related to the business model opportunities (WP6). Originating from at least 6 defined geoclusters (including climatic, economic and social parameters), which aim to be representative of different latitude realities for the housing sector, a selection of building archetypes will be established by AIGUASOL and CARTIF. These building archetypes will consider existing information of reference European projects for Single Family Houses and Multi Family Houses (i.e. Tabula, Entranze or 4RinEU). Initial requested information on archetypes will be completed with other relevant data from official European (Eurostat) and national information sources related to building national regulations, socioeconomic conditions and local user patterns or other relevant references (PlugN-Harvest). From the collected and derived information, a first draft version of a structured repository (geodatabase CARTIF) will be implemented and populated ready to be used in the following tasks.

Task 2.3. Definition of the focused calculation procedures and relevant KPIs for the different circular economy composing vectors (Leader: ITeC; Participants: AIGUASOL, TURNTOO, CARTIF, ALCN, LEITAT) [M5-M18]

The aim of this task is to properly and reliably establish reference calculation methodologies for the different circular aspects, and define its main KPIs. To ensure replicability a set of common core indicators is needed for making environmental, social and economic improvements to be used in decision-making through the entire life-cycle of buildings. To support this development, the task will be aligned to:

- The results from the working groups involved in the common EU framework of indicators for the environmentalperformance of buildings. (i.e. Communication 'Resource Efficiency Opportunities in the Building Sector' COM (2014)
445)
- The LCA methodology as defined in ISO 14040/44 norms and existing software for LCA; The LCC methodology asdefined in ISO 15686-5 and other relevant documents The standards developed/under development by CEN/TC 350 Sustainability of construction works (e.g. EN 15840; EN 15643; EN 15978; EN 16309; EN 16627), based on LCA and LCC methodology for construction products and buildings.
- Product Environmental Footprint (PEF); Leesman and verification tools as KPMG true value; Cost optimalmethodology as established in EPBD recast 2010;
- Level(s) new buildings sustainable methodology developed by the European Commission.

As a minimum, the following issues will be tackled: energy, water and material consumption; durability of construction products; design for deconstruction; CDW; recycled content in construction materials; recyclability and reusability of construction materials and products. ITeC and TURNTOO will focus on defining environmental and economic KPIs related to materials and CDW, whereas LEITAT and CARTIF will define the KPIs focused on water, waste and energy aspects such as life cycle Global Warming Potential (kg CO2 eq), Ozone Depletion Potential (kg CFC eq),water resource depletion (m3 water eq), Water Exploitation index (WEI+), cumulative energy demand (MJ), cost by life cycle stage (€/ m2/yr), etc. ALCN will contribute to define KPIs by researching local framework conditions (local standards, rules and legislation), contacting local and global initiatives concerning circularity, and conducting interviews with experts from ALCN´s network. AIGUASOL will contribute to consider, adapt and define the specific energy calculation procedures and KPIs', and proposing the matching of these with other indicators and procedures related to the other above-mentioned environmental aspects (water and material consumptions, waste, etc.). In task 3.1 the stakeholders will be engaged, it will be at this juncture at the project start that stakeholders will be requested to give inputs into other potential indicators that can be used to measure the social impact (task 5.4) and to give their feedback on the current indicators.

Task 2.4. Integration of the calculation procedures in an overall circular economy methodology (Leader: AIGUASOL; Participants: CARTIF, ITeC, ALCN) [M14-M26]

From the results previously obtained in Task 2.2 (reference data for the building archetypes on the representative geoclusters), and Task 2.3 (main calculation procedures and KPIs for the different vectors composing the circularity of buildings), in Task 2.4 an overall methodology to determine the degree of circularity for residential buildings will be defined. This would be a multi-dimensional model of quantitative as well as qualitative parameters, ready to be used at the initial or decision stages of new or retrofitted building definition. The aim of the Building Circularity Methodology is to establish a clear approach to a calculation method that will offer valuable indicators of the degree of potential circularity business opportunities of a building. By that the BCM will establish a global calculation procedure able to determine the potential of reusability and recyclability of the materials, physical solutions, water and waste, and the energy life cycle, under a common approach and considering potential interrelations. As the first approach to the BCM, the methodology will be implemented by AIGUASOL and CARTIF in some theoretical scenarios defined in Task 2.2 to obtain a first database of results. This database will be used in a mathematical

sensitive analysis to be developed by AIGUASOL, CARTIF and ALCN to establish the weighting factors of the main KPIs aiming to simplify the BCM as much as possible.
Task 2.5. Refinement of the BCM methodology and establishment of the reliability factors on the results [HOUSEFUL Solution: S1] (Leader: AIGUASOL; Participants: CARTIF, AHC, ALCN) [M26-M48] The aim of Task 2.5 is to refine the BCM preliminary version defined in Task 2.4, in a loop process with other direct tasks related to the real demo cases (WP4 by CARTIF), environmental and economic assessment (WP5) and business models (WP6), but also considering oriented feedbacks coming from the follower buildings and benchmarking with other existing or on-going EU projects on the same scope, as well as stakeholder inputs. AIGUASOL will be in charge to define an easy to use template to fill in all the required refinement data to be provided by other partners or third parties. AIGUASOL and CARTIF will also be in charge, jointly with the demo owners and the demo implementation WP leader, to collect these data and, by using the results obtained in Task 2.5, compare it with the first draft version. Homeowners/responsible partners of the frontrunner buildings (AHC and ALCN) will participate in relation to their demo building, providing data and input from the building and feedback on the usefulness and usability of the methodology. From there, a final version of the reliable factors will be introduced, and the overall methodology will be adapted by AIGUASOL and CARTIF. Task 2.6. Existing market circularity: solutions and services (Leader: AIGUASOL; Participants: LEITAT, CARTIF) [M1-M26] Task 2.6 includes a desk research on existing market circularity solutions in order to establish a comprehensive framework of the European current situation. This research will also facilitate the potential engagement of companies that supply these solutions and services, supporting Task 3.1 (identification and analysis of stakeholders). The successful circular solutions and services identified will be included in the SaaS developed in WP6 to promote replication in housing sector.

| Participation per Partner | |
|---|---|
| **Partner number and short name** | **WP2 effort** |
| 1 - LEITAT | |
| 2 - ITeC | |
| 3 - CARTIF | |
| 4 - ALCN | |
| 5 - AHC | |
| 6 - VISUM | |
| 7 - AIGUASOL | |
| 9 - GWR | |
| 10 - Turntoo B.V. | |
| 16 - HOMEBIOGAS LTD | |
| 17 - IRIS | |
| **Total** | |

List of deliverables

| Deliverable Number[14] | Deliverable Title | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|
| D2.1 | Report on end-users requirements and CEBO on residential buildings | 7 - AIGUASOL | Report | Public | 6 |
| D2.2 | Circular economy repository for building archetypes in the geoclusters | 7 - AIGUASOL | Report | Confidential, only for members of the consortium (including the Commission Services) | 12 |
| D2.3 | Reference methodologies and KPIs in circular economy analysis | 7 - AIGUASOL | Report | Public | 18 |

List of deliverables

| Deliverable Number[14] | Deliverable Title | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|
| D2.4 | Existing market circularity solutions and product | 7 - AIGUASOL | Report | Public | 26 |
| D2.5 | First approach of BCM methodology and implementation in theoretical scenarios | 7 - AIGUASOL | Report | Public | 26 |
| D2.6 | Description of the BCM methodology and implementation rules in real scenarios | 7 - AIGUASOL | Report | Public | 48 |

Description of deliverables

Deliverables: D2.1) Report on end-users requirements and CEBO on residential buildings [M6, AIGUASOL] / D2.2) Circular economy repository for building archetypes in the geoclusters [M12, AIGUASOL] / D2.3) Reference methodologies and KPIs in circular economy analysis [M18, AIGUASOL] / D2.4) Existing market circularity solutions and products [M26, AIGUASOL] / D2.5) First approach of BCM methodology and implementation in theoretical scenarios [M26, AIGUASOL] / D2.6) Description of the BCM methodology and implementation rules in real scenarios [M48, AIGUASOL]

D2.1 : Report on end-users requirements and CEBO on residential buildings [6]

Information gathering regarding end-users needs related to the CEBO on residential buildings will be done. This information will be used to define a valuable-by-design service through implementing a Quality Function Deployment methodology, which will be the basis for the development of the rest of the work in WP2. This deliverable is related to Task 2.1.

D2.2 : Circular economy repository for building archetypes in the geoclusters [12]

Definition of the main circular economy conditions of building archetypes for representative European geoclusters by cross-referencing main environmental aspects (energy, water, materials and waste) against at least six of the most representative geoclusters (including climatic, economic and social parameters). This deliverable is related to Task 2.2.

D2.3 : Reference methodologies and KPIs in circular economy analysis [18]

Establishment of the most relevant calculation methodologies and KPIs to be considered in the circular economy analysis to be deployed during the project. The technically pre-selected KPIs will be then considered when engaging stakeholders in task 3.1., as well as for the circular economy baseline evaluation (D4.3) and for the validation of HOUSEFUL solutions at building scale (D4.6). This deliverable is related to Task 2.3.

D2.4 : Existing market circularity solutions and product [26]

Market analysis and prospection looking for available circularity solutions and products, establishing a comprehensive European framework, providing key data and descriptions of each of them, as well as giving support to task 3.1 regarding stakeholders identification and analysis. A coordination between this deliverable and D6.1 economic information regarding market analysis will be promoted. This deliverable is related to Task 2.6.

D2.5 : First approach of BCM methodology and implementation in theoretical scenarios [26]

The BCM will establish a global calculation procedure able to determine the potential of reusability and recyclability of the materials, physical solutions, water and waste, and the energy life cycle, under a common approach and considering potential interrelations. As the first approach to the BCM, the methodology will be implemented in some theoretical scenarios defined in Task 2.2 to obtain a first database of results. This database will be used in a mathematical sensitive analysis to establish the weighting factors of the main KPIs aiming to simplify the BCM as much as possible. This deliverable is related to Task 2.4.

D2.6 : Description of the BCM methodology and implementation rules in real scenarios [48]

Detailed explanation on BCM methodology developed during the project for identifying the degree of potential circularity business opportunities of a building, also considering real scenarios validation, will be developed. This BCM methodology will be fed by all the previous tasks included in WP2 and results from WP4. This deliverable is related to Task 2.4 and Task 2.5.

## Schedule of relevant Milestones

| **Milestone number[18]** | **Milestone title** | **Lead beneficiary** | **Due Date (in months)** | **Means of verification** |
|---|---|---|---|---|
| MS1 | Methodology for evaluating the circularity level of buildings | 7 - AIGUASOL | 26 | D2.5) The methodology defined in WP2 will be validated using real scenarios (all Frontrunner buildings) and integrated in the SaaS developed in WP6. |
| MS2 | Desk research of existing circular solutions & services | 7 - AIGUASOL | 26 | D2.4) Existing market circular solutions |

| Work package number [9] | WP3 | Lead beneficiary [10] | 13 - WE&B |
|---|---|---|---|
| Work package title | Social engagement: starting the co-creation of HOUSEFUL services | | |
| Start month | 1 | End month | 54 |

**Objectives**

WP3 has two objectives: (1) To analyse and map out the stakeholders' belief structures underlying their behaviour, attitudes, and societal concerns in order to address knowledge requirements for the demonstration and replication of the 11 HOUSEFUL solutions exploited as integrated services; 2) To effectively engage local and regional stakeholders from Frontrunner buildings in the co-creation of the proposed 11 HOUSEFUL Services according to their needs and behavioural patterns, making feasible and realistic the co-creation of new circular economy business opportunities (CEBOs) from developed solutions. The methodology used and developed in this WP will be the basis for the demonstration of the HOUSEFUL Solution S2.

Specific Objectives:

- Identifying and mapping the key stakeholders and their relationships in terms of knowledge, influence and interestin the Frontrunner buildings;
- Designing and implementing an effective engagement strategy for the successful co-creation and final definition ofservices offered to stakeholders in each Frontrunner building in order to increase the social awareness and engagement of services co-created for covering their needs.
- Analysing the specific factors influencing behaviour and the choice making structures of people and the potential socialrisks and the barriers affecting the successful demonstration of the proposed HOUSEFUL solutions as new services (identification of the individual and collective needs, definition of common goals and best consensus scenario);
- Implementing an innovative engagement strategy for the successful co-creation and final definition of services offeredto stakeholders in each Frontrunner building in order to increase social awareness and engagement of services co-created for covering their needs.

**Description of work and role of partners**

**WP3 - Social engagement: starting the co-creation of HOUSEFUL services** [Months: 1-54]

**WE&B**, LEITAT, ITeC, CARTIF, ALCN, AHC, AIGUASOL, GWR, Turntoo B.V., Neues Leben, HOUSING EUROPE, ICONS

Task 3.1. Identification and analysis of stakeholders for the co-creation of CEBOs (Leader: WE&B; Participants: TURNTOO, LEITAT, ALCN, AHC, CARTIF, AIGUASOL, ICONS, GWR, NEUES LEBEN) [M1-M42]

The co-creation and demonstration of 11 HOUSEFUL services in selected 4 Frontrunner buildings requires the engagement of different actors forming part of the current linear value chains in the housing sector. The core HOUSEFUL project team (WE&B, TURNTOO, LEITAT, ALCN, AHC, NEUES LEBEN, GWR, CARTIF, AIGUASOL and ICONS) together with the Frontrunner buildings will undertake a stakeholder identification, mapping and analysis of the social values and beliefs of the circularity building models as well as the needs, concerns and expectations of HOUSEFUL services. Some stakeholders will influence the implementation of solutions and services in Frontrunner buildings, others have interest in the project concerning key players at each demo site and identified stakeholders to be engaged. A total of 32 national and international stakeholders have already been identified and have sent a LOI showing their interest in the project and or participating in the CCHE board – these initial stakeholders will aid in snowballing the stakeholders' identification in the context of the frontrunner building and in the international community levels. Local and regional key stakeholders will be identified and mapped through a social network analysis (SNA) method. In combination, a deeper social analysis will be approached by qualitative methods through a series of sectorial interviews and/or focus groups to be undertaken at each of the Frontrunner buildings. This will serve to start identifying insights of stakeholder's behavioural decisions for building circularity (including perceived risks of technologies, materials, etc.) and to answer the question of what drives stakeholders' decision to use CEBO services (attitudes, beliefs and concerns). This will aim to further identify the potential social barriers and enablers to implement the CEBO services. These activities will require an initiation of awareness activities regarding the project that will be coordinated with the communication activities in WP7. Therefore, AHC and NEUES LEBEN will be directly involved in this task to create communication materials for the qualitative analysis implying an initial engagement. With this approach, more qualitative information will be gathered on what the stakeholders could provide to HOUSEFUL, how to effectively engage with them and to better identify their communication channels, etc. The main outputs from this

task will be recommendations and inputs for the final co-creation and replication that will result in an engagement strategy for each building site including the initial list of stakeholders that will be engaged (in task 3.2).

Task 3.2. Engagement and co-creation activities. [HOUSEFUL Solution: S2] (Leader: WE&B; Participants: ICONS, LEITAT, HOUSING EUROPE, AHC, NEUES LEBEN, ALCN, AIGUASOL, ITEC) [M1-M54]
Following the stakeholder analysis defined in Task 3.1, the analysis of social and cultural barriers and enablers from the previous task will follow a process of engagement using social qualitative methods with the identified key stakeholders at each HOUSEFUL Frontrunner buildings. This task, led by WE&B, will also deepen in the analysis of factors underlying the stakeholders' behaviour, values and belief, and attitudes and perceptions towards the use of these buildings. Outputs will be used to define the detailed technical specifications of circular solutions to be demonstrated in WP4.

Subtask 3.2.1 Co-creation workshops: developing a strategic plan for the HOUSEFUL solutions and services (Leader: WE&B) [M1-M54]
The engagement activities will follow a backcasting process with the selected stakeholders, due to its problem-solving character. This process will implement three types of co-creation workshops to start the engagement process in each one of the regions where Frontrunner buildings are located. WE&B will support the leaders of the frontrunners buildings who will be the partners responsible to organise these workshops. The partners involved in this task will support end-users to better understand the implications of incorporating circular solutions in their day-to-day life. The first workshop will be structured on the basis of the identified factors conditioning social behaviour and acceptance. This workshop will aim at identifying and validating barriers and enablers obtained in Task 3.1 but in a participatory approach and co-creating the future scenarios of the social conditions of the CEBOs that stakeholders would desire. The second workshop will be designed based on the results from the first workshop to generate ideas and knowledge for co-creation of communication materials (messages, identification of resources and tools, contents) regarding the services provided for the front-runner buildings. Stakeholders, especially tenants, can ensure that their concerns are addressed and knowledge is generated by co-creating together the contents of future communication of the services. The output of this activity will be the basis for the co-creation of social validated HOUSEFUL services and also further consideration towards behavioural changes after effective engagement (indicators will be framed in the social assessment from WP5). Close to the end of the project, a third workshop will be undertaken with regards to the provision of the feedback and validation of the final services demonstrated in the frontrunner buildings as well as to present the SaaS created in WP6 and to co-create a roadmap for further uptake beyond the project. In parallel to this workshop, an Open House event will be organised as a more technical approach with a professional audience and not only with the concerned community of the frontrunners buildings. The Open House event will be organised in each Frontrunner building to enable a wider community of citizens and stakeholders (not directly involved in the demonstration) to see the buildings' implementations and to share the benefits of the different implemented services. Lead by WE&B, the partners AHC, ALCN and NEUES LEBEN will be involved in this task in relation to the identification and validation of the challenges and opportunities presented by the social housing aspect of the project and the socio-cultural-economic profile of the new tenants and dwellers of the buildings. Collaboration with the local social and/or voluntary social services will be promoted.

| Participation per Partner | |
|---|---|
| **Partner number and short name** | **WP3 effort** |
| 1 - LEITAT | |
| 2 - ITeC | |
| 3 - CARTIF | |
| 4 - ALCN | |
| 5 - AHC | |
| 7 - AIGUASOL | |
| 9 - GWR | |

| | |
|---|---|
| 10 - Turntoo B.V. | |
| 11 - Neues Leben | |
| 12 - HOUSING EUROPE | |
| **Partner number and short name** | **WP3 effort** |
| 13 - WE&B | |
| 14 - ICONS | |
| **Total** | |

List of deliverables

| Deliverable Number[14] | Deliverable Title | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|
| D3.1 | Social engagement strategy for the co-creation of HOUSEFUL solutions as new services (version I) | 13 - WE&B | Report | Public | 18 |
| D3.2 | Social perceptions and beliefs in circular economy at building scale (version I) | 13 - WE&B | Report | Public | 32 |
| D3.3 | Social conditions for the co-creation of all HOUSEFUL services and co-created materials | 13 - WE&B | Report | Public | 48 |
| D3.4 | Social engagement strategy for the co-creation of HOUSEFUL solutions as new services in Demo 2 and Demo 3 | 13 - WE&B | Report | Public | 26 |
| D3.5 | Social perceptions and beliefs in circular economy at building scale (version II - Final) | 13 - WE&B | Report | Public | 42 |
| D3.6 | Social engagement strategy for the co-creation of HOUSEFUL solutions as new services (Final) | 13 - WE&B | Report | Public | 42 |

Description of deliverables

Deliverables: D3.1) Social engagement strategy for the co-creation of HOUSEFUL solutions as new services in Demo 1 and Demo 4 [M18, WE&B including preliminary results from Demo 1 and Demo 4] (final version in M42 including results from Demo 1 once dwellers live in the refurbished building - D3.6) / D3.2) Initial social decisions

and behavior in circular economy at building scale [M32, WE&B, including results from Demo 2, Demo 3, Demo 4] (update in M42 including results from Demo 1 - D3.5) / D3.3) Social conditions for the co-creation of all HOUSEFUL services and co-created materials [M48, WE&B] / D3.4) Social engagement strategy for co-creation of HOUSEFUL solutions as new services in Demo 2 and Demo 3 [M26, WE&B including results from Demo 2 and Demo 3]. / D3.5) Social perceptions and beliefs in circular economy at building scale (version II - Final) [M42, WE&B, including an updated and final version of D3.2 and the results from the Demo 1] / D3.6) Social engagement strategy for the co-creation of HOUSEFUL solutions as new services (Final) [M42, WE&B, Updated and final version of D3.1 and D3.4, including results from Demo 1 once dwellers live in the refurbished building]

D3.1 : Social engagement strategy for the co-creation of HOUSEFUL solutions as new services (version I) [18]

A detailed review on the stakeholders' identification and analysis process for the different CEBOs and DEMO locations will be done. Definition of the social engagement strategy that will assure proper co-creation actions during the project, establishing a clear roadmap for Tasks 3.2, as well as defining the basis for the demonstration of S2 solution. This deliverable is related to Task 3.1. To be updated in M26 including results from Demo 1 (D3.4)

D3.2 : Social perceptions and beliefs in circular economy at building scale (version I) [32]

A report regarding initial social decisions derived from the needs, concerns and expectations of the previously identified stakeholders (Task 3.1) involved in the co-creation process will be presented. This report will be used as an initial comparison reference to be consulted in subsequent stages of HOUSEFUL project, leading to the final demonstration of S2 solution. This deliverable is related to Task 3.2. To be updated in M42 including results from Demo 1 (D3.5)

D3.3 : Social conditions for the co-creation of all HOUSEFUL services and co-created materials [48]

A detailed review of the social and cultural restrains, synergies and drawbacks found during the social engagement process for the co-creation of CEBOs, evaluating the results obtained through the social network analysis and the qualitative methods applied, defining an adequate strategy for demonstrating S2 solution. This deliverable is related to Task 3.2.

D3.4 : Social engagement strategy for the co-creation of HOUSEFUL solutions as new services in Demo 2 and Demo
3 [26]

Updated and final version of D3.1.

D3.5 : Social perceptions and beliefs in circular economy at building scale (version II - Final) [42]

Updated and final version of D3.2.

D3.6 : Social engagement strategy for the co-creation of HOUSEFUL solutions as new services (Final) [42]

Updated and final version of D3.1 and D3.4. It includes results from the Demo 1, finally obtained once dwellers live in the refurbished building

**Schedule of relevant Milestones**

| Milestone number[18] | Milestone title | Lead beneficiary | Due Date (in months) | Means of verification |
|---|---|---|---|---|
| MS3 | Social engagement plan | 13 - WE&B | 26 | Stakeholder analysed process finished for all buildings, related to D3.1) Social engagement strategy. |
| MS4 | Co-created services | 13 - WE&B | 48 | Engagement workshop organised, related to D3.3. |

| | | | |
|---|---|---|---|
| **Work package number [9]** | WP4 | **Lead beneficiary [10]** | 3 - CARTIF |
| **Work package title** | Large scale demo buildings | | |
| **Start month** | 3 | **End month** | 48 |

**Objectives**

This WP aims at the successful demonstration of the circular economy integrated solutions in the four Frontrunner buildings (two buildings located in Spain and two buildings in Austria). In detail, this WP aims to demonstrate the technical feasibility of the technical solutions behind the integrated HOUSEFUL service from material (S3, S4), water (S5, S6), waste (S7, S8, S9) and energy (S10, S11) vectors.

Specific objectives:

- To develop Material Passports of 4 Frontrunner buildings including a detailed inventory of the physical state of eachbuilding and the elements and materials it consists of by using a 3D model (BIM). Demonstration of S5.
- To ensure BIM interoperability along the WP providing all material, element, and system specifications based on thestandard IFC4 (ISO 16739:2013 - Industry Foundation Classes (IFC) for data sharing in the construction and facility management industries).
- To develop the technical specifications of the technological solutions: S6 and S7 for water vector; S8 and S9 for wastevector and S10 and S11 for energy vector.
- To search additional financial resources for proper dimensioning of TS and replication of proposed solutions asservices.
- Selection of companies for renovation works for Frontrunner buildings owned by public entities.
- To manage the required permit for the initiation of demonstration activities in Frontrunner buildings.
- To validate the feasibility of proposed solutions at lab scale before scaling up and demonstration at large scale.

## Description of work and role of partners

**WP4 - Large scale demo buildings** [Months: 3-48]
**CARTIF**, LEITAT, ITeC, ALCN, AHC, AIGUASOL, GWR, Turntoo B.V., Neues Leben, HOUSING EUROPE, IDP, HOMEBIOGAS LTD
The demonstration character of this WP4, indeed, will start in WP3, when engagement of stakeholders is initiated and social and cultural barriers are identified and tackled using activities planned in Task 3.1 and 3.2. The development of the technical project in this WP includes the demonstration of technological solutions in Frontrunner buildings following a simple process: detailed technical specification of all solutions with technological character behind the Integrated Systemic HOUSEFUL Service (Material (S3, S4), Water (S5, S6), Waste (S7, S8, S9) and Energy (S10, S11)). In parallel, permits for technological implementation of proposed solutions in Demo buildings will be sought for approval. From a technical point of view, a strong supervision of the works and commissioning of systems to be installed in Frontrunner Buildings will be carried out for technologies implemented from S5, S6, S7, S8, S10 and S11.

Task 4.1 Circular economy audit: Baseline calculation (Leader: CARTIF; Participants: ITEC, TURNTOO, NEUES LEBEN, ALCN, GWR, AHC, LEITAT) [M1-M48]
In this task is provided an overall and exhaustive circular economy audit of the initial status of the four Frontrunner buildings in all four vectors: water, waste, energy and materials and in four dimensions: technological (LEITAT, CARTIF), social (ALCN, AHC), environmental (LEITAT), economic (ITeC, TURNTOO). A specific subtask will address materials deeply using Material Passports.
The aim is to facilitate, in the next steps, the technical definition, specifications (Task 4.2) and optimisation (Task 4.3) of the HOUSEFUL solutions and the monitoring program (Task4.4) of each demo-case. The audit will take into account the initial set of KPI's and the calculation methods defined in WP2. A specific subtask will address materials deeply using Material Passports. The four Frontrunner buildings will be modelled with BIM and be analysed using Material Passport. Each constructive element of this model will incorporate specific KPIs to analyse the circularity of buildings and facilitate proposals for improvements focused on the building circularity. Details are shown below.

Subtask 4.1.1. Material Passports integrating BIM for baseline definition and potential evaluation of deconstruction scenarios. [HOUSEFUL Solution: S3] (Leader: ITEC; Participants: TURNTOO, CARTIF, NEUES LEBEN, ALCN, GWR, AHC) [M1-M48]

The materials used in all four Frontrunner buildings will be recorded according to the planning schemes. The material passport will be created, as well as a full possible deconstruction plan for after life application. Before beginning the refurbishment of frontrunner buildings, hazardous building materials (e.g. asbestos, lead, mercury, polychlorinated biphenyls (PCB), chlorofluorocarbons, etc) will be identified with the aim to know which items need to be safely removed, segregated and disposed. The materials located in this building will again be calculated through the BIM according to the plans of the building. The material passport will be compiled together with deconstruction scenarios and possible implementation of other innovative service based business models of HOUSEFUL. Each constructive element of this model will incorporate KPIs to analyse the circularity of buildings and facilitate proposals for improvements. ITEC will carry out the BIM model of the 4 Frontrunner buildings, whereas TURNTOO will be in charge of producing a Materials Passports which is a digital document that records the identity of all construction materials used, documenting the products and raw materials used in a building. It also contains information about the quality of the materials and their location. This makes reusing and reclaiming materials during demolition or disassembly much simpler. This information will be specified under the standard IFC to assure the interoperability to include such information in BIM.

Task 4.2 Technical specification of HOUSEFUL solutions (Leader: CARTIF; Participants: LEITAT, ALCN, GWR, TURNTOO, AIGUASOL, ITEC, IDP, HOMEBIOGAS) [M1-M26].

Subtask 4.2.1. Technical specifications for solutions including technological implementation. [HOUSEFUL Solutions: S5, S6, S7, S9, S10 and S11] (Leader: CARTIF; Participants: LEITAT, ALCN, GWR, AIGUASOL, ITEC, IDP, HOMEBIOGAS) [M1-M26].
The main goal of this task is the definition of the technical specification (TS) of each HOUSEFUL solution including the technological implementation in each Frontrunner building considering the principles of the circular economy. KPIs defined in WP2 will be applied in the detailed specifications to crosscheck the strategies for implementation. Each solution will be designed by specific partners with expertise on it, supervised by CARTIF in order to ensure that the circularity of the overall solution remains as a valuable Circular Economy Business Model. Each solution will be specified under the standard IFC to assure the interoperability to include such solutions in the BIM models.

Subtask 4.2.2. Technical specifications for solutions not including technological implementation. [HOUSEFUL Solutions: S4, S9] (Leader: CARTIF; Participants: ITEC, TURNTOO, AHC, GWR, NEUES LEBEN) [M1-M26]. This Subtask defines the technical specifications required for the proper demonstration of HOUSEFUL solutions without including a technological implementation (S4 and S9). In detail for S4, the type, number/quantity, origin an functionalities, among other main characteristics for those building materials available for renovation/refurbishment at local scale will be defined. Regarding S9, the results obtained in previous subtask 4.1.1 (material passport and BIM) will be used as basis information for the development of Design for Disassembly (DfD) evaluation of Frontrunner buildings. Technical specifications regarding constructive systems and materials to allow highly percentage of reusability of these components in other buildings or materials will be defined.

Task 4.3. Optimisation of technological solutions before final implementation in Frontrunner buildings (Leader: LEITAT, Participants, ALCN, IDP, HOMEBIOGAS, CARTIF, ITEC, AHC, NEUES LEBEN, AIGUASOL) [M5-M36] Since specific solutions in Water (S5, S6), Waste (S7, S8) integrate innovative technologies for the treatment of household effluents and BWs, validation tests will be conducted firstly at lab scale (TRL 6) in order to adjust and optimise the proper configuration and operation of proposed systems, and then deployed at full scale in Frontrunner buildings (TRL7-8) in Task 4.8. In detail, the following Subtasks are planned to be executed:

Subtask 4.3.1. Optimisation of Water vector
S5 components (Leader: ALCN, Participants: IDP; LEITAT)

The main goal of this subtask is the optimisation of NBS (as green facade or green roof) and O3/UV technologies for greywater and rainwater treatment for reuse in toilets in Frontrunner buildings 1 and 2. First approach includes the optimisation of NBS at laboratory (TRL5) before scale up (at TRL 7) and integration by ALCN (with the support of IDP in Task 4.8). First tests will include the selection of plants and analysis of their performance according to (Spanish and Austrian) climate conditions on the treatment of selected WW by analysing parameters such as COD, BOD, total Nitrogen, Suspended Solids, among other parameters. ALCN will provide calculations and experimental implementations concerning the size of treatment systems (i.e. m2 per litre influent), retention times, dilution factors and plant consortia. In addition, experiments will be carried out to produce specific parts of the NBS using 3D printing optimising the material use. Once the NBS systems are optimised, they will be scaled up according to water (greywater and rainwater) treatment needs in Frontrunner building 1 ("Mestres" - Sabadell) and 2 ("SANT QUIRZE" – Sant Quirze del Vallès). Potential water losses by evapotranspiration will be considered. The treated and collected water will be used for toilet flushing and irrigation. Then, first samples obtained by ALCN at laboratory scale will be treated by LEITAT at lab scale (TRL6) by using O3/UV technologies. Optimisation tests will include the analysis of O3 dosage, hygienisation time and power for hygienisation as well as microbiological tests such as E. coli or salmonella content. Once NBS are installed in Frontrunner 1 and 2 (Task 4.8), LEITAT will validate the optimal operating conditions of commercial O3/ UV technologies (TRL9) using the same equipments and methods used at laboratory scale. Results will be shared with IDP.in order to select and adjust the proper operation of O3/UV commercial systems installed in Task 4.8.

LIST OF TECHNICAL SPECIFICATIONS
VECTOR: WATER
---
HOUSEFUL Solution: S5. Efficient treatment and reuse of rainwater and greywater streams using green and low-energy consuming solutions
TS 5.1.1: Greywater and blackwater separation (IDP) - DEMO 1: Spain "Els Mestres" – 8 homes.
TS 5.1.2: Greywater and blackwater separation (IDP) - DEMO 2: Spain "SANT QUIRZE" – 4 homes.
TS 5.2.1: Modifying greywater discharge system for treatment and reuse using innovative NBS (ALCN) and commercial O3/UV hygienisation (IDP) - DEMO 1: Spain "Els Mestres" – 8 homes.
TS 5.2.2: Modifying greywater discharge system for treatment and reuse using innovative NBS (ALCN) and commercial O3/UV hygienisation (IDP) - DEMO 2: Spain "SANT QUIRZE" – 4 homes.
TS 5.3.1: Integrating water meters at household and building level: controlling water reuse (IDP) - DEMO 1: Spain "Els Mestres" – 8 homes.
TS 5.3.2: Integrating water meters at household and building level: controlling water reuse (IDP) - DEMO 2: Spain "SANT QUIRZE" – 4 homes.
TS 5.4.1: Rainwater pre-treatment for reuse using NBS on facade or roof (ALCN) and integration in building (IDP, ALCN) - DEMO 1: Spain "Els Mestres" – 8 homes.
TS 5.4.2: Rainwater pre-treatment for reuse using NBS on facade or roof (ALCN) and integration in building (IDP, ALCN) - DEMO 2: Spain "SANT QUIRZE" – 4 homes.
---
HOUSEFUL Solution: S6. Efficient treatment and reuse of rainwater and greywater streams using green and low-energy consuming solutions
TS 6.1: Separation of liquid/solid fraction of wastewater and organic content. Treatment of liquid fraction with NBS indoor, at facade and outdoor - DEMO 3: "CAMBIUM" – 12 homes.
---
VECTOR: WASTE
---
HOUSEFUL Solution: S7. Efficient treatment and valorisation of BW and blackwater for biogas production
TS 7.1: Design and integration of AnMBR system (IDP, LEITAT) - DEMO 1: Spain "Els Mestres" – 8 homes.
TS7.2: Design and Integration of dAD system (HOMEBIOGAS, ALCN) - DEMO 3: "CAMBIUM" – 12 homes.
TS 7.3.1: Biogas cleaning for renewable heat production (CARTIF) - DEMO 1: Spain "Els Mestres" – 8 homes.
TS 7.4.1: Storing cleaned biogas in a safety and compact system. Final valorisation according to co-creation (CARTIF, IDP, ALCN, LEITAT) – DEMO 1: Spain "Els Mestres" – 8 homes.
TS7.4.2: Storing biogas in a safety and compact system. Final valorisation according to co-creation (CARTIF, IDP, ALCN, LEITAT) - DEMO 3: "CAMBIUM" – 12 homes.
---
HOUSEFUL Solution: S8. High quality fertilizer/compost of local origin
TS 8.1: Stabilisation of digestate for final use in local gardening (public parks, training activities in schools and/or neighbourhood). Specifications according to national regulation on compost use (LEITAT, ALCN) - DEMO 1: Spain "Els Mestres" – 8 homes.

TS 8.2: Stabilisation of digestate for final use in local gardening (public parks, training activities in schools and/ or neighbourhood). Specifications according to national regulation on compost use (LEITAT, ALCN) - DEMO 3: "CAMBIUM" – 12 homes.

---

VECTOR: ENERGY

---

HOUSEFUL Solution: S10. Energy efficient facades and roofs using modular and sustainable materials TS10.1: Substituting asbestos façade (ITEC, AHC) - DEMO 2: Spain "SANT QUIRZE" – 24 homes.

TS10.2: Design of an energy-efficient facade using sustainable materials with advanced functionality according to building/ community(dwellers) needs (ITEC, AIGUASOL, AHC) - DEMO 2: Spain "SANT QUIRZE" – 24 homes.

TS10.3.1: Design and improvement of energy-efficient facades using modular and/or more sustainable materials selected under circular criteria (ITEC, AIGUASOL, GWR, ALCN, AHC) - DEMO 1: Spain "Els Mestres" – 8 homes.

TS10.3.2: Design and improvement of energy-efficient facades using modular and/or more sustainable materials selected under circular criteria (ITEC, AIGUASOL, GWR, ALCN, AHC) - DEMO 3: "CAMBIUM" – 12 homes.

---

HOUSEFUL Solution: S11. Guarantee energy savings/ production in the building
TS11.1: Co-creation based on performance rather than ownership for energy systems (i.e. ST, PV) finally installed in each building (Pay per performance business model). Monitoring of energy saving by active and passive measures from material/energy valorisation points of view (AIGUASOL) - DEMO 1: Spain "Els Mestres" – 8 homes.
---

S6 components (Leader: ALCN)
The main goal is the optimisation of NBS indoor (TRL 6), at facade (TRL 5) and outdoor (TRL 6) of liquid fraction of WW with high organic content for water reuse. ALCN will provide calculations and experimental implementations concerning the size of treatment systems (m2 per litre influent), retention times, dilution factors and plant consortia adjusted to different climates. In order to utilise the treated water directly on site it can be reused in toilets and/or irrigation (fertigation for conversion of nutrients in treated water to food). Farming activities will include tests with: reusable materials (e.g. fire hoses or plastic bottles) in vertical and horizontal systems; aquaponics system with the option to collect rainwater and to make use of organic residues for converting into fish feed. Results will allow the proper installation for bigger prototypes, finally integrated into real environment (TRL 7) and possibly serving the building with water cycles.

Subtask 4.3.2 Optimisation of Waste vector
S7 components (Leader: LEITAT, Participants: CARTIF, IDP, ALCN, HOMEBIOGAS)
The main goal is (1) to optimise the operation of the AnMBR system at laboratory scale for the treatment of blackwater and grinded BWs (potentially installed in Frontrunner building 2), and (2) the optimisation of dry anaerobic digestion (dAD) provided by HOMEBIOGAS for the optimal treatment of the solid fraction from non-segregated WW mixed with grinded BW (potentially installed in Frontrunner building 3). In Frontrunner building 1 (Els Mestres building, in Sabadell), biogas cleaning technology based on membrane contactors will be optimised for the efficient removal of CO2, H2S and impurities from biogas in order to adapt the final quality of gas to according to the predefined potential final use (CHP, grid injection, transport, etc.) in the different demos finally defined by co-creation activities.
First, LEITAT will design a compact and modular 5L AnMBR at lab scale. LEITAT will be in charge of its optimisation (biological and membrane systems). Biological reactor will be start-up using an inoculum from existing AnMBR.
Operating parameters such as blackwater:BW ratio, particle size, organic load, hydraulic retention time and cellular residence time will be optimised for optimal COD, Nitrogen, TS and VS removal and biogas production. Commercial ultrafiltration polymeric membranes will be selected for optimal ultrafiltration of treated water from biological reactor. Optimisation of filtration performance (filtration and back-washing cycles) will be performed aiming to minimise membrane fouling and filtration cost (i.e. renovation, maintenance, energy cost). Biological and membrane systems will be connected for continuous operation. Potential reuse of treated water as irrigation water or use for toilets will require hygienisation. For that, LEITAT will performed again the S5 tests using the same lab scale O3/UV technologies and the methods. When laboratory trials are finalised, the system will be scaled up. LEITAT will design and construct a compact and transportable AnMBR unit able to treat the organic load rate of 8-10 inhabitant equivalents (blackwater and grinded BW, in optimal ratio previously reported at lab scale). The design will be addressed for guaranteeing safe operating conditions and minor odour production. Hazard and operability analysis (HAZOP) and odour controls (by gas olfatometry gas and chemical analysis of samples by gas chromatography) will be carried out during the operation of AnMBR for optimisation of final design and odour control. The system will be started up and optimised in LEITAT facilities using real WW (from LEITAT building) and grinded kitchen wastes. Once safety guarantees of the system is ensured (by IDP, LEITAT, CARTIF and AHC) and O+M protocols are prepared, the system will be emptied and prepared for transportation to Demo 1 ("Els Mestres" building) for final integration and operation at building scale (Task 4.8). Regarding dAD, after the separation of the dry and liquid fraction of domestic WW in Demo 3, the dry organic content will be transported via slides to the dry anaerobic digestor, placed close to the sewage pipe end and the burning system. The planned digestion system from HOMEBIOGAS will directly remove hydrogen sulfide and unpleasant odours from the gas outlet. Gas characterisation will be performed by ALCN. The system can either further clean the gas for storing enriched methane in a gas storage balloon to save biogas for the winter times, and/or utilise it directly in cooking stoves. Quality of biogas will be monitored by ALCN with the supervision of HOMEBIOGAS. Twice a year we will empty the sludge from the bottom of the tank with the foreseen outlet and a pumping system directly sending it to the composting unit. The liquid that can be removed permanently will also be processed in the composter (S8 component) and will further be collected as liquid fertiliser. Other interventions with converting waste from Demo 3 could include: the breeding of insects fed with organic waste, which can further be used as fish feed; the collection of plastic waste, shredding, filament making and big size 3D printing for using in prototypes; old furniture upcycling (see also WP7); collection of old fire hoses and utilisation for urban farming and other sources; and other co-created types of resource recovery.

CARTIF will develop an innovative small-scale biogas cleaning system based on (CO2 and H2S) absorption technologies (by gas washing plus activated carbon filters). Then gas chromatography tests will be performed aiming to optimise final gas quality according to operating parameters such as gas flow, temperature and pressure. The system will be scale up to ensure a biogas quality corresponding to natural gas grid quality specification (>95% CH4, water vapour content
<80mg/m3, etc.) for potential grid injection, use in CHP, combustion in a natural gas boiler, cooking stoves or other valorisation routes proposed by stakeholders for co-creation at building scale. Installation of proposed equipments will be performed by CARTIF/IDP in Demo 1 and CARTIF/ALCN in Demo 3.

S8 components (Leader: LEITAT, Participants: ALCN, IDP)
Several technologies for the stabilisation of AnMBR digestate will be tested at laboratory for optimal selection and scale up to large scale (considering as part of decision making the requirements and needs of stakeholders). In detail, final stabilisation techniques for AnMBR digestate such as drying (LEITAT), domestic co-composting by mixture with fresh BWs (LEITAT) and vermicomposting (LEITAT) will be previously studied at laboratory (TRL5) (treating 50 kg maximum per batch and technique) before optimisation (TRL6) and scale-up and demonstration at large scale by IDP (TRL7 in Demo 2). For mentioned technologies, physic-chemical analysis of compost characteristics such as pH, moisture, organic matter, soluble salts, C:N ratio, ammonia content, biodegradation, odour, maturity (Rottegrade index) and density will be performed and compared. Hygienisation degree will be also analysed (E. coli and Salmonella sp.) by LEITAT. On the other hand, a compost cultivator prototype will be developed by ALCN at TRL6 and optimised for composting fresh gardening wastes before mixing with dAD digestate at large scale in Demo 3. Optimisation of optimal ratio will be performed according to compost characteristics evaluated by ALCN. For rainwater-fed NBS, a compost leaching system can be considered to encourage composting and enable use of green walls also with water low in nutrients.

Subtask 4.3.3 Optimisation of Energy vector
S10 components (Leader: ITEC, Participants: AHC, AIGUASOL, NEUES LEBEN, ALCN, HOUSING EUROPE)
The energy optimisation has to be tackled from two perspectives: active and passive/bioclimatic. The active system needs energy consumption but some active systems need less energy than others. Otherwise, passive systems don't need energy and became more efficient but sometimes is difficult to apply it in a residential refurbishment.
AIGUASOL and ITEC will work in the feasibility analysis of innovative active and passive systems potentially installed as improved elements in building structure in all Frontrunner buildings. Some examples of this active and passive systems improving the energy efficiency of buildings may include modular facades, roofs, cross ventilation, galleries, awnings, conditioning systems for radiant surfaces, evaporative cooling through water or vegetation films, Trombe walls, Canadian wells, and other solutions including energy generation from the building, among others. Co-creation between AIGUASOL and ITEC with stakeholders (including AHC, ALCN and NEUES LEBEN) will promote the design and potential installation of optimal systems to be selected according to circular criteria as well as their cost (installation, maintenance and renovation cost). The final installation of selected objects in Demo 1 and 2 will require the study of new financial resources to be searched in Task 4.6 (lead by HOUSING EUROPE). ITEC will be the partner in charge of the modular façade definition and AIGUASOL will help to define the energy properties of the developed modular façade. To do so, it will be used the EPBD established cost optimal procedure in order to check the best options to be developed and implemented and considering different scopes (cost optimal buildings and nZEB solutions).

S11 components (Leader: AIGUASOL, Participants: AHC, ALCN)
Co-creation opportunities based on performance rather than ownership for energy systems (i.e. Solar Thermal (ST) and Photovoltaic systems (PV)) will be evaluated to be implemented in Frontrunner buildings. Pay per performance business models applied to renewable energy systems will be promoted. Under this model, an energy service company supplies, installs and maintains a ST or PV system, guaranteeing an annual solar production defined in a contract of 10 years. By using this model, it is guaranteed that ST or PV systems operate correctly at least during the contract signed between the energy service company and the owner/neighbourhood community of the building. Renting or leasing options will also be considered when implementing renewable energies.

Task 4.4 Technical monitoring program (Leader: ALCN; Participants: CARTIF, LEITAT, IDP, AIGUASOL, HOMEBIOGAS) [M21-M48]
According to the technologies and solutions selected in the previous tasks, a monitoring program will be specified to ensure a correct impact assessment of the demonstration. Therefore, suitable sensors will be selected (photometric sensors, online microbial sensors, humidity sensors, temperature sensors, etc.).
Regarding solutions designed for WW treatment (S5 and S6), ALCN will include sensors for the evaluation of water quality after NBS treatment. Water meters will be installed by IDP for quantification of water reuse in toilets at home and building level after O3/UV treatment in Frontrunner building 1 and 2. Regarding Solution 7, partners involved will integrate sensors and monitoring required for the operation of AnMBR system foreseen in Frontrunner building

1. CARTIF will proceed with a simultaneous real-time monitoring of CH4, H2S and CO2 after the cleaning stage in order to confirm that the gas stream fulfils the quality requirements demanded. CH4, H2S will be measured by dual wavelength infrared cell with reference channel, while H2S will be measure by electrochemical cell. The data acquisition and the repository of data according to the KPIs will be done in cloud-based systems to oversee the data sets online at any time. The data monitoring can be linked to automation protocols to increase for example retention time of any treatment automatically if a threshold is exceeded. The cloud-based system could further provide the capability to link the static information from the BIM server (IFC compliant) to the dynamic data repository (BACN standards compliant) to show all detailed information by means of a dynamic BIM visualisation tool.

Task 4.5 Execution project documents (Leader: CARTIF; Participants: ALCN, LEITAT, IDP, AIGUASOL, ITEC, HOMEBIOGAS) [M1-M36]
Once the technical specification of the solutions in frontrunner buildings (Task 4.2) and the associated monitoring elements are defined (Task 4.3), the execution project documents will be produced. The documents will also be supported and enriched by the Material Passports and BIM models of each building (Task 4.1). This will serve to the following two tasks: generation of the financial scheme (Task 4.6) and after that the launching of the tender process (Task 4.7).

Task 4.6 Improved access to finance through investment planning (Leader: HOUSING EUROPE; Participants: ALCN, AHC, AIGUASOL, ITEC, LEITAT) [M1-M48]
Further than the EC contribution, other financial resources to create a feasible investment plan for proper dimensioning of technical specifications as well as for allowing the replication of 11 HOUSEFUL solutions marketable as services will be sought. There are several options that will be explored: Other national or regional funding schemes, RIS3, crowd funding through ethical banks, bank loans and/or guarantees through blending instruments from EIB (e.g. InnovFin, ESIF) and national banks, ERDF (if applicable). It is also foreseen to explore other alternative financial schemes, mainly those that will use as foundations an innovative public and private business models or even an only private model. The task includes the development of a financing plan, the progress monitoring of the implementation of the plan for each Frontrunner building, a risk/uncertainty assessment, mitigation plans and the evaluation of different potential scenarios and impacts for the project. New economic resources promoting the final definition and exploitation of HOUSEFUL solutions as services will be welcome and offered for successful replication activities in Follower buildings or new case studies after the end of project execution.

Task 4.7 Selection of companies for renovation works (Leader: AHC; Participants: IDP, ALCN, GWR, LEITAT, AIGUASOL, ITEC) [M18-M30]
Since the two Spanish Frontrunner buildings are owned by a public entity, a tender process will be necessary to the development of the works. For “Els Mestres” building, due to the immediate renovation works of the building, the tendering process and selection of companies will be initiated before HOUSEFUL execution. For “SANT QUIRZE building”, the tendering and selection of companies will be executed once the standard AHC refurbishment project is approved . Therefore, in this last case, tendering documents will be prepared. AHC with the technical support of IDP, AIGUASOL, ITEC and LEITAT, will be in charge of this process, according to the Spanish rules and normative for public procurement processes. Within the other Austrian demo site the companies will be selected according to price and trust. ALCN with the support of GWR and “CAMBIUM” residents will be in charge of this selection process.

Task 4.8 Management of permits, implementation and supervision of technological solutions at building scale (Leader: IDP; Participants: ALCN, GWR, AHC, AIGUASOL, ITEC, LEITAT, CARTIF, HOMEBIOGAS) [M6-M48]
In this task, the company or companies selected will implement the different solutions and monitoring elements in the building. Previously, all permits for technological implementation of proposed solutions in each Demo buildings will be sought for approval. Once approved, ALCN in Austria and IDP in Spain will be in charge the technical assistance and the supervision of the works related to the technological innovation shown in S5, S6, S7, S8 and S9. ITEC and AIGUASOL will be in charge of the technical assistance and supervision of works for solutions potentially installed in S10, evaluating the resulted energy performance in S11. Mentioned partners will periodically report to the consortium the progresses and potential delays or difficulties. These partners will follow the commissioning process of the system installed to ensure that the technical solutions are performing correctly. After this phase, the demonstration will be ready to start the environmental and socioeconomic impact assessment (WP5) of the solutions.
A detailed description of partner's role in WP4 has been included in each partner profile included in Part B, Section 4, in the description of the partner role in HOUSEFUL project.

| Participation per Partner | |
|---|---|
| **Partner number and short name** | **WP4 effort** |

| | |
|---|---|
| 1 - LEITAT | |
| 2 - ITeC | |
| **Partner number and short name** | **WP4 effort** |
| 3 - CARTIF | |
| 4 - ALCN | |
| 5 - AHC | |
| 7 - AIGUASOL | |
| 9 - GWR | |
| 10 - Turntoo B.V. | |
| 11 - Neues Leben | |
| 12 - HOUSING EUROPE | |
| 15 - IDP | |
| 16 - HOMEBIOGAS LTD | |
| **Total** | |

List of deliverables

| **Deliverable Number[14]** | **Deliverable Title** | **Lead beneficiary** | **Type[15]** | **Dissemination level[16]** | **Due Date (in months)[17]** |
|---|---|---|---|---|---|
| D4.1 | Circular economy audit: Baseline calculation for Demo 1 and 4 | 3 - CARTIF | Report | Public | 18 |
| D4.2 | Technical specifications and execution project of solutions in Spanish and Austrian demos: Demo 1 and Demo 4 | 3 - CARTIF | Demonstrator | Confidential, only for members of the consortium (including the Commission Services) | 18 |
| D4.3 | BIM models and material passport of the Spanish and Austrian demos: Demo 1 and Demo 4 | 2 - ITeC | Report | Public | 18 |

| | | | | | |
|---|---|---|---|---|---|
| D4.4 | Validation tests of technologies in S5, S6, S7 and S8 solutions at lab scale. Includes the Safety and monitoring plan for S7 and S8 | 1 - LEITAT | Report | Public | 36 |
| D4.5 | Report on potential improvement in financial schemes and funding opportunities for Demo 2 Front-runner buildings and replication activities | 12 - HOUSING EUROPE | Report | Confidential, only for members of the consortium (including the Commission Services) | 26 |
| D4.6 | Report on potential improvement in financial schemes and funding opportunities for Demo 3 Front-runner buildings and replication activities | 12 - HOUSING EUROPE | Report | Confidential, only for members of the consortium (including the Commission Services) | 30 |

List of deliverables

| Deliverable Number[14] | Deliverable Title | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|
| D4.7 | Report of demonstration and implementation of HOUSEFUL solutions at building scale. Includes the Technical monitoring programme | 3 - CARTIF | Demonstrator | Public | 48 |
| D4.8 | Report on financial schemes and funding opportunities for all Front-runner buildings and replication activities (final update) | 12 - HOUSING EUROPE | Report | Confidential, only for members of the consortium (including the Commission Services) | 48 |
| D4.9 | Technical specifications and execution project of solutions in Spanish and Austrian demos (final version) | 3 - CARTIF | Demonstrator | Confidential, only for members of the consortium (including the Commission Services) | 36 |
| D4.10 | BIM models and Material passport for the Spanish and Austrian demos integrating HOUSEFUL solutions | 2 - ITeC | Report | Public | 48 |
| D4.11 | Circular economy audit: Baseline calculation for Demo 2 and Demo 3 | 3 - CARTIF | Report | Public | 23 |

| | | | | | |
|---|---|---|---|---|---|
| D4.12 | Technical specifications and execution project of solutions in Spanish and Austrian demos: Demo 2 and Demo 3 | 3 - CARTIF | Demonstrator | Confidential, only for members of the consortium (including the Commission Services) | 26 |
| D4.13 | BIM models and material passport of the Spanish and Austrian demos: Demo 2 and Demo 3 | 2 - ITeC | Report | Confidential, only for members of the consortium (including the Commission Services) | 30 |

| Description of deliverables |
|---|
| Deliverables: D4.1) Circular economy audit: Baseline calculation for Demo 1 and Demo 4 [M18, CARTIF] D4.2) Technical specifications and execution project of solutions in Spanish and Austrian demos: Demo 1 and Demo 4 |

[M18, CARTIF] / D4.3) BIM models and material passport of the Spanish and Austrian demos: Demo 1 and Demo 4 [M18, ITEC] / D4.4) Validation tests of technologies in S5, S6, S7 and S8 solutions at lab scale. Includes the Safety and monitoring plan for S7 and S8 [M36, LEITAT] / D4.5) Report on potential improvement in financial schemes and funding opportunities for Demo 2 Frontrunner buildings and replication activities. It includes a financing plan, progress monitoring of the implementation of the plan, risk/ uncertainty assessment, mitigation plans and different scenarios and impacts for the project. [M26, HOUSING EUROPE] / D4.6. Report on potential improvement in financial schemes and funding opportunities for Demo 3 Front-runner buildings and replication activities [M30] [HOUSING EUROPE] / D4.7) Report of demonstration and implementation of HOUSEFUL solutions at building scale. Includes the Technical monitoring programme [M48, CARTIF] / D4.8. Report on financial schemes and funding opportunities for all Frontrunner buildings and replication activities (updated and final version of D4.5 and D4.6) [M48] [HOUSING EUROPE] / D4.9) Updated and final version of D4.2 and D4.12 [M36, CARTIF] / D4.10) BIM models and material passport of the Spanish and Austrian demos integrating HOUSEFUL solutions [M48, ITeC]. / D4.11) Circular economy audit: Baseline calculation for Demo 2 and Demo 3 [M23, CARTIF] / D4.12) Technical specifications and execution project of solutions in Spanish and Austrian demos: Demo 2 and Demo 3 [M26, CARTIF] / D4.13) BIM models and material passport of the Spanish and Austrian demos: Demo 2 and Demo 3 [M30, ITEC]

D4.1 : Circular economy audit: Baseline calculation for Demo 1 and 4 [18]

This Deliverable reports the results of an exhaustive circular economy audit on the initial status of two Frontrunner buildings, Demo 1 and Demo 4, in the four targeted vectors (i.e. water, waste, energy, materials) and in four dimensions (i.e. technological, social, environmental, economic), considering the identified KPIs and calculation methods set in WP2 (D2.3). The baseline data will be a key element for the technical definition specifications (task 4.2), optimization (task 4.3) and monitoring (task 4.4). This deliverable is related to Task 4.1.

D4.2 : Technical specifications and execution project of solutions in Spanish and Austrian demos: Demo 1 and Demo 4 [18]

This deliverable compiles the results from two interrelated HOUSEFUL activities: Task 4.2 and Task 4.5, related with Demo 1 and Demo 4. Consequently, the D4.2 will be divided in main two parts that will include: (1) Technical specifications of solutions including both technological implementation (S5, S7, S10 and S11) and (2) those not including technological implementation (S4, S9) for the Frontrunner buildings 1 and 4 (Task 4.2). A first version of D4.2 including this part for mentioned buildings (Demo 1 and Demo 4) will be delivered in M18; (2) The execution projects will be related with the technological solutions (Task 4.5) and delivered as an updated and unique version of D4.2 and D4.12 in M36 for the 4 Frontrunner buildings (D4.9). The execution project will be completed once results from optimization tests (at laboratory scale) are obtained and sensors/control system required by the monitoring program are initially defined in Task 4.4. Both technical specification and the execution projects will be supported by the BIM models and material passport of Demo 1 and Demo 4 generated in Task 4.1 and the co-creation process promoted during WP3.

D4.3 : BIM models and material passport of the Spanish and Austrian demos: Demo 1 and Demo 4 [18]

The D4.3 will present a detailed review of the inventory of used materials of the Frontrunner buildings 1 and 4 considering both material passport strategies, the definition of deconstruction plans for afterlife applications and the BIM modelling for the demo sites. This task relates to the establishment of HOUSEFUL solution S3 for mentioned buildings but will be also used for the definition of technical specifications and execution projects in Task (D4.2). This deliverable is related to Task 4.1.

D4.4 : Validation tests of technologies in S5, S6, S7 and S8 solutions at lab scale. Includes the Safe-ty and monitoring plan for S7 and S8 [36]

The D4.4 summarizes the results from the laboratory tests deployed for the validation of S5 (rainwater and greywater treatment with NBS and O3/UV commercial system), S6 (unsegregated water treatment with NBS and O3/UV commercial system), S7 (blackwater and biowaste treatment for biogas production through AnMBR, dAD and absorption solution for biogas cleaning), and S8 (stabilized biowaste from AnMBR ad dAD used for producing high quality fertilizer/ compost). A safety and monitoring plan for S7 and S8 solutions will be included. This deliverable is related to Task 4.3 (Subtask 4.3.1 and Subtask 4.3.2).

D4.5 : Report on potential improvement in financial schemes and funding opportunities for Demo 2 Front-runner buildings and replication activities [26]

The main goal of this deliverable is to report the strategy for the identification of financial opportunities for the co-financing of refurbishment works in Frontrunner buildings, as well as for the promotion and increase in the implementation of HOUSEFUL solutions in more homes per building. The search of co-financing will include funding from private and public origins. Local and national alternatives in Austria and Spain, as well as general European options, will be considered. For that, a Financing plan will be delivered in M26. The report shall

include a progress monitoring plan for each co-funding opportunity (identified), a detailed risk/uncertainty assessment, mitigation plans and different scenarios and impacts. This deliverable shall include a detailed financing plan for Demo 2. This deliverable is related to Task 4.6.

D4.6 : Report on potential improvement in financial schemes and funding opportunities for Demo 3 Front-runner buildings and replication activities [30]

The main goal of this deliverable is to report the strategy followed for the search of funding opportunities for both the co-financing of refurbishment works in Demo 3 Frontrunner building, as well as for the promotion and increase in the implementation of HOUSEFUL solutions in other buildings. The search of co-financing will include funding from private and public origin. The report shall include a progress monitoring plan for each co-funding opportunity (identified), a detailed risk/uncertainty assessment, mitigation plans and different scenarios and impacts. This deliverable is related to Task 4.6.

D4.7 : Report of demonstration and implementation of HOUSEFUL solutions at building scale. Includes the Technical monitoring programme [48]

A detailed report regarding results during demonstration stage of HOUSEFUL technological solutions (S5, S6, S7, S8, S9, S10 and S11) in the Frontrunner buildings will be presented. These results will be directly related to the selected KPIs of D2.3, providing data for the environmental, economic and social assessment of HOUSEFUL solutions (WP5) and the subsequent exploitation and business models analyses of WP6. The deliverable will be divided into 3 main parts which will include: 1) Management of permits (Subtask 4.8.1). All permits for the technological implementation of HOUSEFUL solutions will be identified, managed and sought for approval in each Frontrunner building. For that, technical specifications for each HOUSEFUL solution in each Frontrunner building will be the basis for the management of these permits, considering specific risks, ethics concerns and project milestones. It is worth to mentioning that "Els Mestres" building (Demo 1) has initiated the procedure for their refurbishment before the HOUSEFUL kick off. The refurbishment includes some of the HOUSEFUL solutions (S3, S4 and S10). Then, management of some permits has been initiated for the successful demonstration of S3, S4 and S10 during the refurbishment. 2) Implementation (Subtask 4.8.2): the report will include a description of the procedure used for the installation, start-up and optimization of HOUSEFUL solutions in each Frontrunner building (S5, S6, S7, S8, S9, S10 and S11), as well as and the quantification of KPIs (defined in D2.3) demonstrating the achievement of project results. 3) Supervision of technological solutions Subtask 4.8.3 and Task 4.4): Includes the Technical monitoring program of the project. This deliverable is related to Task 4.8.

D4.8 : Report on financial schemes and funding opportunities for all Front-runner buildings and replication activities (final update) [48]

This deliverable will include an updated versions of D4.5 and D4.6 reports, which describe the financial schemes and funding opportunities searched for all Demos. This deliverable is related to Task 4.6.

D4.9 : Technical specifications and execution project of solutions in Spanish and Austrian demos (final version) [36]
Updated and final version of D4.2

D4.10 : BIM models and Material passport for the Spanish and Austrian demos integrating HOUSEFUL solutions [48]

Updated and final version of D4.3 and D4.13 including HOUSEFUL solutions in the four Frontrunner buildings.

D4.11 : Circular economy audit: Baseline calculation for Demo 2 and Demo 3 [23]

Following the same methodology used for the development of D4.1, this Deliverable reports the results of an exhaustive circular economy audit on the initial status of the Frontrunner buildings 2 and 3 in the four targeted vectors (i.e. water, waste, energy, materials) and in four dimensions (i.e. technological, social, environmental, economic), considering the identified KPIs and calculation methods set in WP2 (D2.3). The baseline data will be also a key element for the technical definition specifications (task 4.2), optimization (task 4.3) and monitoring (task 4.4). This deliverable is also related to Task 4.1.

D4.12 : Technical specifications and execution project of solutions in Spanish and Austrian demos: Demo 2 and Demo 3 [26]

Using the same approach of the D4.2, this deliverable also compiles the results from two interrelated HOUSEFUL activities: Task 4.2 and Task 4.5, related with Demo 2 and Demo 3. Consequently, the D4.2 will be divided in main two parts that will include: (1) Technical specifications of solutions including both technological implementation (S5,
S6, S7, S10 and S11) and those not including technological implementation (S4, S9) for the Frontrunner buildings 2 and 3 (Task 4.2). D4.12, including this part for mentioned buildings (Demo 2 and Demo 3), will be delivered in M26; (2) The execution projects will be related with the technological solutions (Task 4.5) and delivered as an updated and unique version of D4.2 and D4.12 in M36 for the 4 Frontrunner buildings (D4.9). The execution project will be completed once results from optimization tests (at laboratory scale) are obtained and sensors/control system required by the monitoring program are initially defined in Task 4.4. Both technical specification and the execution projects will be supported by the BIM models and material passport of Demo 2 and 3 generated in Task 4.1 and the co-creation process promoted during WP3.

D4.13 : BIM models and material passport of the Spanish and Austrian demos: Demo 2 and Demo 3 [30]

The D4.13 will present a detailed review of the inventory of used materials of the Frontrunner buildings 2 and 3 considering both material passport strategies, the definition of deconstruction plans for afterlife applications and the BIM modelling for the demo sites. This task relates to the establishment of HOUSEFUL solution S3 for mentioned buildings but will be also used for the definition of technical specifications and execution projects in Task (D4.2). This deliverable is related to Task 4.1.

**Schedule of relevant Milestones**

| Milestone number[18] | Milestone title | Lead beneficiary | Due Date (in months) | Means of verification |
|---|---|---|---|---|
| MS5 | Baseline of Frontrunner buildings completed | 3 - CARTIF | 23 | D4.1) D4.11) Baseline calculation for 4 Frontrunner buildings completed |
| MS6 | Execution project documents ready | 3 - CARTIF | 36 | D4.9) Technical specifications and execution projects of Frontrunner buildings |
| MS7 | Technological solutions ready to be validated at demonstration scale | 3 - CARTIF | 36 | D4.4) Optimisation of technological solutions (Task 4.3), Financial scheme (task 4.6), and technical monitoring programme (task 4.8). |
| MS14 | Financial plan for Demo 2 | 3 - CARTIF | 26 | In task 4.6, delivery of D4.5 due on M26 and updated version on M48 (D4.8). It includes a Financing plan, a progress monitoring of the implementation of the plan, risk/uncertainty assessment and mitigation plans. |
| MS15 | Financial Close for Demo 3 | 3 - CARTIF | 30 | Complementary financial access for demo 3 is completed. Associated deliverables - D4.6 (in M30) and updated version D4.8 (M48). |
| MS16 | Tenants occupy the "Els Mestres" building | 3 - CARTIF | 40 | They begin to participate in co-creation activities (D3.5) |

| | | | |
|---|---|---|---|
| **Work package number [9]** | WP5 | **Lead beneficiary [10]** | 1 - LEITAT |
| **Work package title** | Environmental, economic and social assessment | | |
| **Start month** | 8 | **End month** | 54 |

**Objectives**

- To define the HOUSEFUL assessment plan for the three pillars: environmental, economic and social.
- To assess the potential environmental and socioeconomic impacts of the circular strategies demonstrated under a LifeCycle Thinking approach.
- To determine the energy savings, water reuse, material recycling and waste valorisation potential achieved andassociated to the operating phase of buildings by adopting circular economy principles.

• To test the Level(s) tool developed by the European Commission and provide recommendations for improvement.

## Description of work and role of partners

**WP5 - Environmental, economic and social assessment** [Months: 8-54]
**LEITAT**, ITeC, CARTIF, ALCN, AHC, AIGUASOL, LGI, GWR, Turntoo B.V., Neues Leben, WE&B, ICONS, IDP, HOMEBIOGAS LTD
Task 5.1. HOUSEFUL assessment plan (Leader: CARTIF; Participants: LEITAT, ITeC, AIGUASOL, WE&B) [M13M17] This task aims to deploy the assessment plan for the three pillars of the project: envi-ronmental, economic and social. The assessment plan will be tailor-made to the characteristics of each Frontrunner building. The first step before the assessment of each pillar in Task 5.2, Task 5.3 and Task 5.4 is to define the assessment plan. This task should be strongly aligned with both Task 4.3 (technical monitoring program) because the monitoring program should cover the requirements of the plan and Task 2.3 (defini-tion of the focused calculation procedures and relevant KPIs) since the KPIs should allow to evaluate the pillars of the project and take them into account during the development of the assessment plan. The HOUSEFUL assessment plan covers three subtasks in order to define the plan for each pillar (environmental, economic and social). The environmental pillar will be assessed in two different ways: 1) by applying a life cycle thinking approach considering energy, materials, waste and water impacts (task 5.1.1) and 2) fo-cusing only on the energy impact associated to the operating phase (task 5.1.2), since this represents the highest environmental burden in the life cycle of a building as revealed in LCA studies.

Subtask 5.1.1 Environmental and economic assessment plan (Leader: LEITAT)
The aim of this subtask is to specify the methodology to evaluate the environmental and economic impact of the demonstration activities by applying a life cycle thinking approach. Thus, the boundaries and guidelines for the different stages (production, construction/refurbishment, operation, maintenance and demolition) will be determined and the environmental and economic indicators (aligned with task 2.3) will be selected.

Subtask 5.1.2 Energy assessment plan (Leader: CARTIF) The energy assessment of energy conservation measures will be done under International Measurement and Verification protocol (IPMVP). IPMVP defines the plan to evaluate the energy savings. Thus, in this subtask a Measurement and Verification (M&V) plan will be carried out, consisting in identifying the independent variables, selection of the IPMVP option, base-line energy consumption and definition of the mathematical model. The aim of this plan is to specify the situa-tion of buildings before of interventions.

Subtask 5.1.3 Social assessment plan (Leader: WE&B). The aim is to define the social assessment plan based on behavioural and social criteria (attitudes, values, beliefs and norms). Social assessment procedures (indicators and data gathering processes) will be created on the basis of results from Task 3.1, taking into account the characteristics of each building and owner. The output of this subtask will be a plan for social assessment data gathering that will show the social impact of the project.

Task 5.2. Environmental & economic assessment (Leader: LEITAT; Participants: ITeC, GWR, LGI, AIGUASOL) [M18M54]

Subtask 5.2.1 BIM-LCA&LCC assessment (Leader: LEITAT). The environmental and economic impact of the circular solutions demonstrated in Frontrunner buildings will be assessed in this task based on the assessment plan defined in task 5.1 and the BIM results obtained in WP4. Since the BIM objects from the ITeC's database integrate LCA related information, the environmental impact of the solutions will be assessed from the data obtained from BIM. In line with the guidelines defined by the European Platform on LCA, different impact categories will be considered for determining the environmental impact. In particular, BIM results will generate a set of 22 impact categories (e.g. depletion potential of the stratospheric ozone layer, acidification potential of soil and water, human toxicity potential, eutrophication potential, % of recycled material, hazardous waste, energy of fossil origin, etc). The impact associated to hazardous substances will be assessed by applying a life cycle approach. Hazardous waste can be found in construction and demolition waste. For example, old styrofoam insulation from buildings is considered hazardous waste in Germany or asbestos in old facades is also hazardous. According to the harmonised classification and labelling (CLP00) approved by the European Union, asbestos may cause cancer and causes damage to organs through prolonged or repeated exposure. Potentially hazardous substances should also never be used in circular construction, as the reuse or recycling options are very limited due to health constraints. Some impact assessment categories included in LCA (e.g. hazardous substances, human toxicity) will be used to assess the potential impacts of hazardous substances. GWR and AIGUASOL will contribute to this task by providing relevant/new data for calculating BIM-LCA related information in modified circular buildings. The methodology developed in WP2 to quantify the circularity level of buildings will be tested with data obtained from frontrunner buildings. LEITAT will analyse the results obtained from the impact categories for each demo site in order to compare the environmental impact of different scenarios. Therefore, the environmental impact of the baseline scenario will be compared with the Houseful scenario. A Life Cycle Cost (LCC) of the lifetime cost associated with the buildings' life cycle will be conducted by LGI from the BIM results with the aim to evaluate the economic feasibility of the Houseful model proposed in each Demo Site. Capital costs (design and project management, material costs, construction/refurbishment costs), O&M costs and end of life costs will be taken into account in the LCC process.

Subtask 5.2.2 Level(s) assessment (Leader: LEITAT)
The new Level(s) tool developed by the European Commission will be used to assess the environmental performance of frontrunner buildings. The Level(s) tool will be ready for testing in autumn 2017 and there will be a test period that will last for 2 years. Therefore, the aim of this task is to test Level(s) with real cases (Frontrunner buildingsand provide feedback and recommendations to the European Commission for improving the tool.

Task 5.3. Measurement & Verification of the energy savings (Leader: LEITAT; involved partners: GWR, AIGUASOL,
CARTIF, NEUES LEBEN) [M18-M54]
A Measurement & Verification (M&V) process based on the IPMVP and the energy assessment plan defined in task 5.1.2 will be carried out to determine the energy savings achieved in the frontrunner buildings after the renovation process. Significant variables that affect the energy consumption will be considered such as weather conditions. IPMVP provides 4 options for determining energy savings (A, B, C and D). The options selected for each Demo are outlined below:

Demo 1. Building of the 1960's in Sabadell (Spain)
Option D. Calibrated simulation. Savings will be determined through simulation of the energy performance of the whole building.
Metered energy data for the baseline does not exist. A package of energy conservation measures are expected to be implemented, including envelope interventions.

Demo 2. Building of the 2002's in Sant Quirze del Vallès (Spain)
Option C. Energy savings will be determined by measuring energy consumption at the whole facility utility meter level Metered energy data for the baseline exists. Energy efficiency measures will be implemented, including envelope interventions

Demo 3. Cambium Community Center (Austria)
Option D. Calibrated simulation. Savings will be determined through simulation of the energy performance of the whole building.
Metered energy data for the baseline exists. Energy efficiency measures will be implemented.

Demo 4. New building in Vienna (Austria)
Option D. Calibrated simulation. Savings will be determined through simulation of the energy performance of the whole building.
New constructed building with brief baseline data. Theoretical energy calculations for every part of the building exist

Certified Measurement & Verification Professionals (CMVP) from LEITAT and AIGUASOL will assure that the M&V process of the energy savings is conducted in compliance with the IPMVP. LEITAT will be in charge of comparing the pre and post refurbishment scenarios of demos 2&3 following the option C of the IPMVP, whereas AIGUASOL will be in charge of carrying out the energy simulations (Option D of the IP-MVP) of demos 1&4. GWR and NEUES LEBEN will provide the technical data requested by LEITAT and AIGUASOL.

Task 5.4. Social assessment (Leader: WE&B; Participants: ICONS, AHC, ALCN, NEUES LEBEN) [M18-M54]
To measure the social impact of the HOUSEFUL services, it will be important to not only engage with the HOUSEFUL stakeholders from the very beginning of the project (task 3.1.) but to continually engage with them as the project continues. Feedback from the end users (especially tenants) will be vital to not only tailor the solutions to their needs but to know if these solutions have had an impact on their daily lives. The task will work closely with WP3 and more specifically task 3.1 on stakeholders' identification and analysis and task 3.2 on the engagement of stakeholders.
The social indicators defined within task 5.1 will be assessed with the aim to reveal the final impact of the social aspects on the local and regional population wellbeing and engagement. First, a baseline measurement of these indicators will be undertaken by WE&B, this will serve as a reference point to the future measurements and an indication of the potential positive social impact of the HOUSEFUL project.
The measurement of the social and behavioural indicators will be strategically planned in Task 51, although we can envisage the use of the stakeholder consultation through surveys and interviews at two different scales, the stakeholders at the frontrunner building (including tenants) and at the level of the circularity building international community.In this sense, WE&B will run a survey to a wider audience and stakeholders at the start of the project, its midpoint and again towards the end of the project.
The measurement of the social impact at the start and mid-point of the project will allow for changes and adjustments to be made in the project activities to ensure that the public and end users are satisfied with the use of the solutions and the changes in behaviour they would need to make to adjust to a circular economy.
The project partners will combine both qualitative and quantitative measures to not only get feedback on social insights and behavioural patterns but to also to have numbers and statistics to show and compare the social impact from the start of the project throughout it and again at the end. ALCN will co-organise the stakeholder communication and will react concerning the design of NBS in alignment with the results of stakeholder processes concerning expectations, no-go criteria, etc. NEUES LEBEN as housing cooperative will subcontract this task to their social building support organisations.

| Participation per Partner | |
|---|---|
| **Partner number and short name** | **WP5 effort** |
| 1 - LEITAT | |
| 2 - ITeC | |
| 3 - CARTIF | |
| 4 - ALCN | |
| 5 - AHC | |
| 7 - AIGUASOL | |
| 8 - LGI | |
| 9 - GWR | |
| 10 - Turntoo B.V. | |
| 11 - Neues Leben | |

| | |
|---|---|
| 13 - WE&B | |
| 14 - ICONS | |
| 15 - IDP | |
| 16 - HOMEBIOGAS LTD | |
| **Total** | |

List of deliverables

| Deliverable Number[14] | Deliverable Title | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|
| D5.1 | HOUSEFUL assessment plan | 3 - CARTIF | Report | Public | 17 |
| D5.2 | Environmental and economic impact assessment | 1 - LEITAT | Report | Public | 54 |
| D5.3 | Measurement &Verification | 1 - LEITAT | Report | Public | 54 |
| D5.4 | Social assessment | 13 - WE&B | Report | Public | 54 |

Description of deliverables

Deliverables: D5.1) Houseful assessment plan [M17, CARTIF] / D5.2) Environmental and economic impact assessment [M54, LEITAT] / D5.3) Measurement &Verification [M54, LEITAT] / D5.4) Social assessment [M54, WE&B]

D5.1 : HOUSEFUL assessment plan [17]

An assessment plan will be developed with the aim to define the principles and the framework to be considered for assessing the impact of the circular solutions implemented in each frontrunner building. The assessment plan will present the methodology to assess the impact of the solutions at three different levels: 1) environmental and economics; 2) energy; and 3) social. This deliverable is related to Task 5.1.

D5.2 : Environmental and economic impact assess-ment [54]

This report will present the environmental and economic benefits achieved after implementing the circular solutions in frontrunner buildings. Quantitative results of the environmental impact assessment will be presented. The results of the Life Cycle Cost (LCC) analysis will be presented with the aim to evaluate the economic feasibility of the HOUSEFUL model proposed in each frontruner building. This deliverable is related to Task 5.2.

D5.3 : Measurement &Verification [54]

A Measurement and Verification (M&V) report will be produced presenting the energy savings achieved after refurbishment processes in frontrunner buildings. The report will provide details about the IPMVP option used as well as the independent variables considered for conducting the energy assessment in each demo building. This deliverable is related to Task 5.3.

D5.4 : Social assessment [54]

A social assessment report will be prepared to present the social impact results of the circular solutions implemented in demo buildings. The results of the social and behavioural indicators assessed will be presented with the aim to reveal the final impact of the social aspects on stakeholders, and especially from tenants. This deliverable is related to Task 5.4.

Schedule of relevant Milestones

| Milestone number[18] | Milestone title | Lead beneficiary | Due Date (in months) | Means of verification |
|---|---|---|---|---|
| MS8 | Environmental, economic and social assessment of the circular solutions demonstrated | 1 - LEITAT | 54 | D5.2) Environmental and economic impact assessment; D5.3) Measurement &Verification; D5.4) Social assessment |

| | | | |
|---|---|---|---|
| **Work package number [9]** | WP6 | **Lead beneficiary [10]** | 8 - LGI |
| **Work package title** | HOUSEFUL Solutions as Services, market, policies and exploitation | | |
| **Start month** | 1 | **End month** | 54 |

**Objectives**

The main objective of WP6 is to prepare the market entry strategy of the most promising technological and nontechnological HOUSEFUL solutions as systemic services (including the SaaS) to facilitate the transition from a demonstration phase to a commercialisation phase (T6.2). First, a market study (T6.1) will identify, characterise and quantify most promising markets for the HOUSEFUL services, SaaS included. The development of the HOUSEFUL SaaS (T6.3) will provide a user-friendly interface for the exploitation of the building circularity calculator as well as for the replication of HOUSEFUL solutions as services co-created in new buildings. The business modelling (T6.4.1) and replicability study (T6.4.2) will analyse the impact of the HOUSEFUL services on existing linear business models of the housing value chain stakeholders and identify drivers/barriers to the transition to circular business models. The exploitation strategy (T6.5) will draw on all the collected information to provide the future HOUSEFUL service providers with the most adapted approach to ensure market penetration. Finally, policy recommendations (T6.6) will be done to foster the circularity of the housing sector.

Specific objectives:

- To ensure the marketability of HOUSEFUL solutions as systemic services from the very beginning of the project• To develop a SaaS to encourage the housing value chain to facilitate the transition from linear to circular business models, through the creation of a Circularity Tool (CT) and an interactive sharing platform.
- To deliver a market study highlighting promising markets for the most mature HOUSEFUL services
- To analyse the impact of HOUSEFUL services on existing business models
- To study the replicability of new circular business models under different conditions.
- To propose an exploitation strategy to ease the market entry of most promising and mature HOUSEFUL services• To deliver policy recommendations to foster the circularity of the housing sector

**Description of work and role of partners**

**WP6 - HOUSEFUL Solutions as Services, market, policies and exploitation** [Months: 1-54]
**LGI**, LEITAT, ITeC, CARTIF, ALCN, AHC, AIGUASOL, GWR, Turntoo B.V., Neues Leben, HOUSING EUROPE, WE&B, ICONS, IDP, HOMEBIOGAS LTD, IRIS
Task 6.1: Market study (Leader: LGI; Participants: AHC, NEUES LEBEN, HOUSING EUROPE) [M1-M24]
The objective of the market study is to identify, characterise and quantify promising markets for the most mature and promising HOUSEFUL services and list local businesses and stakeholders along the housing value chain that may be interested in using them. The study will mainly focus on Spain and Austria (where the four demo-sites are located) with the support of the two building owners, respectively AHC and NEUES LEBEN. A preliminary market study and discussions with the project partners, notably HOUSING EUROPE, will be performed to define a strategic scope extension towards other EU countries (Ireland, Belgium, The Netherland, France, etc.). For each geographic market a volumetric estimation of future projects will be performed with a particular attention for the public social housing. Markets will be characterised thanks to common key indicators such as the type of building (private or social), size or the construction date (non-exhaustive list) to identify most promising market. Furthermore, an analysis of existing and foreseen policy and regulation will be performed to complete the study. Finally identified markets will be compared one to another to highlight the most promising markets. Focusing on this shortlist of markets the study will identify local businesses and stakeholders that may be interested in implementing the HOUSEFUL services. Local businesses

working along the housing value chain, from equipment providers to building owners, will be identified and characterised. The established list could be used considered during further replication activities.

Task 6.2: From Solutions to Services. Strategy for a complete co-creation (Leader: LEITAT; Participants: all) [M1-M54] The main goal of this task is to guarantee the successful co-creation, implementation and demonstration of proposed HOUSEFUL solutions as new systemic services. From a detailed monitoring of all project activities (in WP2, WP3, WP4, WP5) affecting the concrete realisation of each solution, the identification market constrains and opportunities for each solution (Task 6.2) and the detailed definition of the exploitation plan for all solutions as potential services (Task 6.4),

LEITAT will work to ensure that the effective co-creation of CEBO between stakeholders is feasible. First, for each HOUSEFUL solution, a Solution-to-Service Monitoring Plan (SSM Plan) will be developed. Each SSM Plan will include the following information: (1) Calendar including the main key steps to be achieve/achieved during project execution affecting the materialisation of each solution to offer as a service; (2) Key milestones including potential constrains during the demonstration phase; and (3) Detailed contingency plan. The SSM Plan will be created by LEITAT with the collaboration of all project partners at the beginning of project execution during first 3 months and will be updated every year, considering all outputs from all HOUSEFUL work packages and activities. When all key milestones in the SSM plan are successfully overcome and market study and exploitation plan for each solution are defined, detailed exploitation strategy and supply costs of each systemic services will be defined for their first potential replication in 10 additional Follower Buildings selected during project execution (minimum amount of buildings to be proposed by CCHE Board in WP7) or new ones from dissemination activities proposed in HOUSEFUL.

Task 6.3: Development of the HOUSEFUL SaaS and functionalities (Leader: VISUM, IRIS; Participants: all) [M8-M54] A SaaS will be developed to support stakeholders (construction companies, technological and product suppliers, design teams, municipalities, regional/national housing agencies, owners, dwellers, tenants and policy makers) to bring residential buildings to the circular economy using solutions offered as services in HOUSEFUL. The SaaS will be embedded it into an ergonomic and tailored graphic user interface (GUI) in order to facilitate its use and exploitation. The GUI will be web-based and device responsive in order to allow the access on mobility via different devices. Since different types of users are expected to access to the HOUSEFUL service, different types of user accounts will be created to allow proper scalability of functionalities and personalisation according to the need of the user. The SaaS will be composed of two main modules: 1) a Circularity Tool (CT) to quantify the circularity level of residential buildings. The algorithm for the calculation of building circularity developed in WP2 is the cornerstone of the HOUSEFUL project as it allows a diagnostic of existing buildings before performing any action. Users will be able to work with data and calculation methods at two defined levels: a) a first level (free access-freemium business model) that will provide the simplest way to assess the circularity level of buildings based on few simple data and key KPIs and b) a second level with advanced functionalities (a pay per use) for professionals (i.e. design teams, construction companies, etc) that wish to assess the circularity level in a more accurate and reliable way, suggesting also different types of interventions that can be applied to improve it. It will be also possible to quantify the effect of new interventions or refurbishment on an existing building in order to choose the most suitable. In both levels, the GUI will allow the user to choose in which phase of the building lifecycle apply the simulation and will guide him through a step process in which the input and calculation options have to be provided to obtain the expected output. The HOUSEFUL service will help the user throughout the process suggesting, whenever possible, precalculated values for the unknown parameters; 2) an interactive sharing platform to promote the co-creation of new CEBOs. The platform will contain successful circular solutions and services, the ones demonstrated in Frontrunner buildings (WP4) but also other successful experiences collected in WP2 (task 2.6), with the aim to promote the replication of these initiatives and facilitate the collaboration among stakeholders. This section of the GUI will be based on a dynamic mapping where dynamic filters can be applied to show only the results of interest. The SaaS will help the housing value chain work on their value proposition, encouraging the transition from linear to circular business models. The SaaS will be developed by VISUM with the technical contributions from all partners. Detailed functionalities of the SaaS will be defined by all the consortium during project execution.

Task 6.4: Business modelling and replicability study (Leader: LGI; Participants: TURNTOO, ITeC, WE&B, CARTIF) [M24-M48]
Globally the emergence of circular economy services leads to a fundamental transformation within impacted sectors as traditional linear business models of the stakeholders are progressively reviewed in favour of circular business models.

Subtask 6.4.1: Business modelling (TURNTOO, ITeC, LGI, WE&B, CARTIF) [M24-M36]
This task aims at analysing the specific impacts of HOUSEFUL solutions as services on the business models of local stakeholders involved in the demonstration sites of the project (frontrunner buildings).
Taking advantage of the workshops organised with local communities in the WP3, business models will be described using the Business Model Canvas (Alexander Osterwalder, 2011). Major interactions and relations between different stakeholders as well as main business characteristics as value proposition, revenues sources or cost structure (nonexhaustive list) will be analysed in a transparent and visual way.
The multi-level perspective (Geels, 2002) will be used as a basis for the evaluation of the disruptive impact of the HOUSEFUL services on existing business models. Inputs from services design team (WP3 and WP4) and from the market study (T6.4) will be used to identify key drivers and barriers to the deployment of the HOUSEFUL services within the existing, considering current business models as the regime and HOUSEFUL services as innovation niches. Selected tools recommended by the Climate KIC`s Climate Change Adaptation Innovation Approach will be considered (based on Climate KIC Pioneers toolkit, 2017). The involvement in the task of the WP3 (WE&B) and

WP4 (CARTIF) leaders will facilitate interactions with local stakeholders necessary to successfully perform the analysis.

Subtask 6.4.2: Replicability study (ITeC, TURNTOO, LGI) [M36-M48]

The longevity and sustainability of new circular business models have to be assessed in a broader context and from different points of view. This task will use the market information of this WP, preliminary results of the co-creation (WP3) and demonstration activities (WP4) as well as the lifecycle analysis from WP5 to assess the direct replicability of the HOUSEFUL services. An important contribution from the CCHE Board is expected for the description of the Follower buildings. Key success factors for replication as well as barriers will be identified in order to assess the replicability potential of the most promising and mature HOUSEFUL services. The outputs of this task will be used to write the exploitation plans (T6.5), the policy recommendations to foster the circularity of the housing sector (T6.6) and will be communicated during the replication oriented-dissemination in Follower buildings (T7.4.1). A roadmap including the necessary actions needed to ensure a successful replicability in Europe will also be included in this task.

Task 6.5: Exploitation Plan (Leader: LGI; Participants: all partners) [M1-M54]

This task aims at gathering all the information produced along the project to draft a market entry strategy. It will consider inputs from non-technical services (WP3), technical demonstration (WP4), environmental and socio-economic assessment (WP5), market analysis (WP6) to define the most adequate approach to a successful exploitation for the most promising HOUSEFUL services. Innovative methodologies will be used facilitate strategic discussions and overcome the barriers created by the multi-actor and multi-resource approach of the project. A LEGO© Serious Play© workshop with LGI certified facilitators will be organised within the first year of the project to better consider the different point of views, define a common vision and draft the first exploitation axes. Other methodologies such as Value Proposition Design (Alexander Osterwalder, 2011) may be used all along the project depending on encountered barriers.

The Exploitation task will consider different levels of activities, in particular:

- Evaluation of possible exploitation outcomes of demonstration results of the project
- Identification of IP right needs, monitoring and support on individual and joint actions
- Specific business model recommendations for each service
- Recommendation on effective exploitation strategies
- Economic and business assessment when necessary
- Additional general recommendations on technology management, financial engineering and market roadmap.The final plan shall cover all project related technologies & services that have been identified as mature enough and promising by the consortium.

Task 6.6 Policy recommendations to foster the circularity of housing sector (Leader: HOUSING EUROPE; Participants: LEITAT, AHC, NEUES LEBEN) [M48-M54]

The aim of this task consists in providing policy recommendations at local, regional, national and EU scale to increase the circularity level of the housing sector in our mainly linear economy. Pushing the supply and demand side by themselves is not sufficient to achieve a circular economy, and specific policies and regulations for the housing sector should be set to achieve a full circular transition.

Existing policies that promote renewable energies and energy efficiency in buildings need to be further strengthened and complemented with policies for resource efficiency, which look at a wider range of environmental impacts across the lifecycle of buildings. HOUSING EUROPE with the support of partners involved in this task will set EU recommendations on how to scale-up circular economy solutions and services in housing sector, including social housing. Legislation and regulatory barriers identified in WP3, during the market study (T6.2) and lessons learned during the project will be used to provide policy recommendations and strengthen the legal framework towards a circular economy. A better use of existing policy instruments will be proposed to promote circular procurement and the launch of circular tendering processes for public social housing which will scale-up new business models.

Task 6.7: Intellectual Property Rights management (Leader: LGI; Participants: All partners) [M1-M54]

This task aims to monitor and manage IPR related decisions derived from HOUSEFUL innovations. In close consultation with WP leaders, LGI as Innovation Manager will: i) Raise awareness of HOUSEFUL consortium members to the IPR rules set up in the Consortium Agreement that regulate the communication and exploitation of project results, in addition to the IPR provisions in the Grant Agreement and Rules for Participation of Horizon2020; ii) Ensure that the communication and exploitation activities comply with the IPR publication policies; iii) Assist the moderation of possible IPR related conflicts; iv) Monitor potential patent applications or other copyright related issues, being WP leaders responsible for the identification, monitoring and dissemination of the project outcomes (know-how, products, semi-products) and iv) Monitor and prepare for potential knowledge transfers between partners or external providers during or after the project duration. All actions developed during this task and subtasks will be described and explained in Exploitation and Business Plan (D6.7).

| |
|---|
| |

**Participation per Partner**

| Partner number and short name | WP6 effort |
|---|---|
| 1 - LEITAT | |
| 2 - ITeC | |
| 3 - CARTIF | |
| 4 - ALCN | |
| 5 - AHC | |
| 7 - AIGUASOL | |
| 8 - LGI | |
| 9 - GWR | |
| 10 - Turntoo B.V. | |
| 11 - Neues Leben | |
| 12 - HOUSING EUROPE | |
| 13 - WE&B | |
| 14 - ICONS | |
| 15 - IDP | |
| 16 - HOMEBIOGAS LTD | |
| 17 - IRIS | |
| **Total** | |

**List of deliverables**

| Deliverable Number[14] | Deliverable Title | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|
| D6.1 | Market study (version I) | 8 - LGI | Report | Public | 18 |
| D6.2 | Solution-to-Service monitoring plan (version I) | 1 - LEITAT | Report | Confidential, only for members of the consortium (including the Commission | 18 |

| | | | | | |
|---|---|---|---|---|---|
| | | | | Services) | |
| D6.3 | HOUSEFUL SaaS | 17 - IRIS | Other | Confidential, only for members of the consortium (including the Commission Services) | 36 |
| D6.4 | HOUSEFUL SaaS – Final version | 17 - IRIS | Other | Public | 54 |
| D6.5 | Impact study of HOUSEFUL services on linear business models | 10 - Turntoo B.V. | Report | Public | 36 |
| D6.6 | Preliminary Exploitation Plan | 8 - LGI | Report | Confidential, only for members of the consortium (including | 36 |

List of deliverables

| Deliverable Number[14] | Deliverable Title | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|
| | | | | the Commission Services) | |
| D6.7 | Replicability analysis of new CEBOs | 2 - ITeC | Report | Confidential, only for members of the consortium (including the Commission Services) | 48 |
| D6.8 | Final Exploitation Plan | 8 - LGI | Report | Confidential, only for members of the consortium (including the Commission Services) | 54 |
| D6.9 | Policy recommendations | 12 - HOUSING EUROPE | Report | Public | 54 |
| D6.10 | Market study (version II - final) | 8 - LGI | Report | Public | 24 |
| D6.11 | Solution-to-Service monitoring plan (version II) | 1 - LEITAT | Report | Confidential, only for members of the consortium (including the Commission Services) | 36 |

| | | | | | |
|---|---|---|---|---|---|
| D6.12 | Solution-to-Service monitoring plan (version III - final) | 1 - LEITAT | Report | Confidential, only for members of the consortium (including the Commission Services) | 51 |

**Description of deliverables**

Deliverables: D6.1) Market study [LGI, M18] [Updated version: M24 - D6.10] / D6.2) Solution-to-Service monitoring plan [LEITAT, M18] / D6.3) HOUSEFUL SaaS [IRIS, M36]. / D6.4) HOUSEFUL SaaS - Final version with completed functionality [IRIS, M54] / D6.5) Impact study of HOUSEFUL services on linear business models [Turntoo, M36] / D6.6) Preliminary Exploitation Plan [LGI, M36] / D6.7) Replicability analysis of new CEBOs [ITEC, M48] / D6.8) Final Exploitation Plan [LGI, M54] / D6.9) Policy recommendations [HOUSING EUROPE, M54] / D6.10) Updated version of D6.1 [LGI, M24] / D6.11) Updated version of D6.2 [LEITAT, M36] / D6.12) Updated and final version of D6.11 [LEITAT, M51]

D6.1 : Market study (version I) [18]

Detailed identification, characterization and quantification of most promising markets for the HOUSEFUL services, SaaS included (D6.3). Most important local business and stakeholders who could be interested in implementing HOUSEFUL solutions will be listed, coordinating with task 3.1. Furthermore, a strategic scope extension to other EU territories considering key indicators and future project volumetric estimations will be calculated. This deliverable is related to Task 6.1.

D6.2 : Solution-to-Service monitoring plan (version I) [18]

Definition of a Solution to Service Monitoring Plan (SSM Plan) for each of the HOUSEFUL services implemented. These SSM Plans will include a detailed calendar with the main key steps to be achieved, the key milestones and constrains during demonstration phase, and a detailed contingency plan. Through them, adequate co-creation strategies oriented to establish exploitable circular economy services will be guaranteed, supported by the Exploitation Plan definition of D6.4 and D6.7. This deliverable is related to Task 6.2.

D6.3 : HOUSEFUL SaaS [36]

A SaaS to promote the engagement of stakeholders to bring HOUSEFUL solutions to implementation will be developed. It will be based on two separated modules: a Circularity Tool to quantify the circularity level of residential buildings, based on the work deployed in WP2; and an interactive sharing platform to promote co-creation of new CEBO, where successful circular solutions and services will be included. This SaaS will consider different stakeholders typologies, allowing a broad degree of personalization. The first version of the HOUSEFUL SaaS will be delivered in Month 36. This deliverable is related to Task 6.3.

D6.4 : HOUSEFUL SaaS – Final version [54]

Delivery of the final version of the SaaS, with open access to all public at Month 54. This deliverable is related to Task 6.3.

D6.5 : Impact study of HOUSEFUL services on linear business models [36]

Establishment of the impact of circular solutions on local stakeholders linear business models by Business Model Canvas, multilevel perspective and Climate Change Adaptation Innovation Approach. Synergies among D6.4, WP3 workshops and WP4 tasks will be capital to properly reach the local stakeholders. This deliverable is related to Task 6.4.

D6.6 : Preliminary Exploitation Plan [36]

Gathering of all the information obtained throughout the first 3 years of the project to set a preliminary draft of a market entry strategy. HOUSEFUL services data (WP3, WP4, WP5) and information related to market (D6.1) will be considered. First results regarding strategic discussion will be presented. This deliverable is related to Task 6.5.

D6.7 : Replicability analysis of new CEBOs [48]

Evaluation of the replicability of HOUSEFUL new CEBOs by analyzing the data generated in the project, mainly the market information of D6.1, preliminary results of the co-creation (WP3) and demonstration activities (WP4), and the lifecycle analysis from WP5; as well as the expert insight on the description of following buildings provided by the CCHE Board. Key success factors for replication and barriers will be identified and replicability potential for all HOUSEFUL services will be established. The outputs of this task will be used for exploitation plans (D6.5 and D6.7) and policy recommendations (D6.8), being also communicated during the replication oriented-dissemination in Follower buildings (T7.3.1). This deliverable is related to Task 6.4.

D6.8 : Final Exploitation Plan [54]

A definitive document establishing a final exploitation plan for each of the HOUSEFUL services, based on the previous work done for D6.5 together with more detailed discussion with each of the technological and service partners. Each HOUSEFUL service exploitation plan will include an evaluation of project outcomes, an identification of IP right needs, monitoring and support on individual and joint actions, specific business model recommendations, a proposal on effective exploitation strategies, and information regarding other economic and business assessment (if required). This deliverable is related to Task 6.5.

D6.9 : Policy recommendations [54]

Report on policy recommendations at local, regional, national and EU scale to promote the transition to circular models for the housing sector based on HOUSEFUL case study experience and data (WP3, WP4, WP5). Information regarding legislation and regulatory barriers identified in WP3 and D6.1 will also be integrated. This deliverable is related to Task 6.6.

D6.10 : Market study (version II - final) [24]

Updated and final version of D6.1.

D6.11 : Solution-to-Service monitoring plan (version II) [36]

Updated version of D6.2.

D6.12 : Solution-to-Service monitoring plan (version III - final) [51]

Updated and final version of D6.11 (being D6.11 an updated version of D6.2).

**Schedule of relevant Milestones**

| Milestone number[18] | Milestone title | Lead beneficiary | Due Date (in months) | Means of verification |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| MS9 | Launch of the market study | 8 - LGI | 24 | D6.1) Market study |
| MS10 | Launch of the SaaS | 8 - LGI | 36 | D6.3) SaaS ready to be used by stakeholders |
| MS11 | Complete list of potential HOUSEFUL services for stakeholders/end-users | 8 - LGI | 54 | D6.7) Final Exploitation Plan |

| Work package number [9] | WP7 | Lead beneficiary [10] | 14 - ICONS |
|---|---|---|---|
| Work package title | Communication and Dissemination | | |
| Start month | 1 | End month | 54 |

**Objectives**

The core objective of WP7 is to manage and implement impact-based dissemination, communication and engagement activities with the aim to increase public and stakeholders' awareness, stimulate acceptance and foster replication of HOUSEFUL services from Frontrunner buildings towards Followers as well as supporting further uptake at European level and beyond. An integrated impact-driven approach will be adopted through a multi-stakeholder and multi-channel strategy, with the following specific objectives:

- To develop a Dissemination and Communication (D&C) Plan to design and manage an effective D&C strategyimplementation at EU and national level as well as guarantee public and professional coverage to enable widespread replication of the project's outcomes;
- To increase public awareness on the activities and the achievements of the project through the development of aneffective public communication strategy based on a multi-channel approach also exploiting HOUSEFUL partners' own communication channels;
- To build and engage with the HOUSEFUL community, implementing dedicated networking activities addressingEuropean citizens and stakeholders and replication-oriented actions addressing the Followers, through the support of the CCHE board;
- To measure the impacts generated by the implementation of D&C activities, through the intelligent use of outreachdata, quantitative performance indicators and key qualitative assessments;

**Description of work and role of partners**

**WP7 - Communication and Dissemination** [Months: 1-54]
**ICONS**, LEITAT, ITeC, CARTIF, ALCN, AHC, VISUM, AIGUASOL, LGI, GWR, Turntoo B.V., Neues Leben, HOUSING EUROPE, WE&B, IDP, HOMEBIOGAS LTD, IRIS

Task 7.1. Communication and dissemination strategy. (Leader: ICONS; Participants: LEITAT, WE&B, HOUSING EUROPE, AHC) [M1-M54]

A Dissemination and Communication Plan will be delivered at the beginning of the project (M6). The main purpose of the plan will be to strengthen the vision and results of the project on a large scale based on a well-defined, impactbased communication strategy to support acceptance and replication at local level (involving both the players of the frontrunner buildings and the cities and EU level. The outreach of and engagement with HOUSEFUL contents will be continuously monitored to measure the effectiveness of the D&C strategy and to guarantee the achievement of the expected impacts. The Plan will outline targets, activities, channels, tools and timing. More specifically it will:

- Identify the target audiences, with the aim to build up a network of stakeholders representing the potential adoptersof the proposed HOUSEFUL solutions and methodology.
- Define key messages. At the beginning of the project a set of key messages will be developed to ensure the HOUSEFULconcepts are received by the target audiences.
- Define communication and dissemination formats, actions, channels and plan of activities.

Task 7.2. Communication materials for co-creation activities with stakeholders in Frontrunner buildings (Leader: ICONS; Participants: All) [M1-M48].

In detail, at local level, the following specific communication materials will be developed to guarantee the engagement of stakeholders for co-creation activities in WP3:

- A call-to-action (CTA) video, a specific video format based on animations and infographics aimed at involving thetarget audience to undertake an action, will be produced to support citizen engagement activities. The CTA will be produced with English voiceover and translated into local languages (Spanish and German) by the Local communication task groups.
- Flyers will be produced with simple and direct messages to inform and engage local citizens about the HOUSEFULsolutions demonstrated into the project, including the SaaS. The flyer will have a common visual identity, but contents may be customised according to the specificity of each Frontrunner building.
- A social media strategy definition, including identification of appropriate channels, messages and specific hashtagsto foster engagement of stakeholders.

Task 7.3. Awareness raising activities and HOUSEFUL identity (Leader: ICONS; Participants: all) [M1-M54] Subtask 7.3.1 – Visual Identity (ICONS): The project visual identity will be implemented at the beginning of the project (M2), including the project logo, graphic elements and templates for presentations and reporting. A flyer will be produced in M10 to inform relevant target groups about the project, its objectives and expected impacts. A certain number of copies will be delivered to the partners for wide distribution at fairs, conferences and workshops. To ensure local outreach, the Local communication task groups will translate the flyer in local language.
A project web video will be produced at M8 to present the project, its objectives and the circular economy concepts in an easy-to-understand way based on animations and info-graphics, for online distribution via the website, social media, communication portals and platforms.

Subtask 7.3.2 – Website and social media (ICONS, all for content provision). The HOUSEFUL website (in English) will be designed and launched within the first five months from the project's start and it will be used as the primary online communication channel and interface towards different target audiences. Before the launch of the HOUSEFUL website, a simple project landing page will be online (M2) reflecting its visual identity, EU funding references and contact details. Direct link to the SaaS developed in WP6 (Task 6.3) for stakeholders and citizens will be established. In view of supporting local communication and engagement, dedicated sections will be developed in local languages, including updates and local news from the Local communication task groups towards local audiences, stimulating acceptance and engagement. Cross-linking and referencing with other external platforms and initiatives will be supported to increase clustering activities. The project website will be maintained for at least 24 months after the end of the project. Social media will be exploited as an additional communication, dissemination and engagement channel and as a participatory tool to foster dialogue, enhance public understanding, engagement and acceptance of stakeholders and citizens. A specific social media strategy will be developed to assess the most appropriate social media channels for HOUSEFUL.

Task 7.4 Replication-oriented dissemination (Leader: ICONS; Participants: all) [M3-54] A two level dissemination strategy will be implemented for replication purposes:

Subtask 7.4.1 Replication between Frontrunner and Follower buildings (Leader: HOUSING EUROPE; LEITAT; ICONS; Local Task Groups).
The Collaborative Community of Housing Experts (CCHE) will engage 10 additional demo-sites for replication of the proposed solutions in Europe. Throughout the execution of the project, dissemination activities and formats will be tailored in view of fostering networking and replication from the Frontrunners to the Followers. Specific peer-to-peer learning and knowledge-sharing tools and activities will be set up: 1. dedicated replication-oriented contents, packaged in info-packs, periodic newsletters, detailed technical descriptions 2. Capacity-building activities will consist in the organisation of periodic webinars, where the Frontrunner buildings will update followers on the project progress and results, and two training sessions and study tours (one in Barcelona and another one in Vienna) organised by the Local Task Groups.
The 10 Follower buildings will be selected according to different criteria: 1) common in EU building archetypes; 2) located in different climatic conditions; 3) different socio-cultural-economic profile of tenants; and 4) interest and commitment of building owners to replicate the circular principles demonstrated in HOUSEFUL frontrunner buildings. Follower buildings can undertake activities that complement and expand related project activities and these could be done in the same timeline and/or beyond HOUSEFUL. In any case the investments associated to the implementation of circular solutions in the follower buildings would be out of the scope of the project and therefore not funded within HOUSEFUL. However, HOUSEFUL will inform Follower buildings about funding and financing options for replicating the solutions demonstrated in the frontrunner buildings. HOUSEFUL will test and implement innovative circular solutions in frontrunner buildings, but with the scope of paving the way for the follower buildings to replicate successful solutions later on in their cities for their own benefit.

Subtask 7.4.2 Dissemination towards a broader stakeholder community (Leader: ICONS; Participants: all) to spread know-how and good practices in other EU cities and foster knowledge, replication and uptake of the project results. The potentially interested target groups identified as part of the D&C Plan (also taking into account the inputs from Task 3.1 on stakeholders' identification and the organisations who have already signed a LOI) and representing the potential adopters and replicators of the HOUSEFUL services together with locally identified important stakeholder groups will be approached through dedicated dissemination activities. The presence of EU and national housing associations in the partnership will maximise the HOUSEFUL results' dissemination potential through their existing communication channels and tools and addressing their communities and networks beyond the ones that will be approached via the project dissemination activities. In addition, the members of the CCHE board will also act as ambassadors of the project concepts and results towards their communities: The Dissemination tools will include:

- e-Newsletter - a periodic e-Newsletter will be issued every six months and distributed to the stakeholders' database ofregistered users. It will be produced with innovative self-publishing tools enabling content publishing and aggregation from multiple external sources in the format of an interactive and easy-to-read e-zine.
- At least 6 info-packs (on both the specific areas: water, waste, energy and material resources and the implementedsolutions) will be produced and distributed to registered web users and key stakeholders and circulated through the partners' channels and promoted via social media;
- Dissemination webinars – four webinars will be organised on the four vectors and the related HOUSEFUL solutions(water, waste, energy and material resources).
- HOUSEFUL Best practice book - in electronic and printable format, containing the main project achievements,is foreseen at the end of the project, providing information about the results, best practices, lessons learnt and recommendations derived by the implementation and demonstration of the HOUSEFUL services in the Frontrunner buildings as well as the challenges tackled. It will be widely promoted towards the HOUSEFUL community and CCHE board.
- Scientific and technical publications dedicated journals as well as conference papers will be performed by academicand technological partners. The objective will be to increase peer-to-peer dissemination and knowledge sharing among expert stakeholders in the circular economy area and to position the project al-so in professional and scientific journals and conferences.
- Targeted workshops and training sessions to the broader community to learn and teach about a variety of circularityaspects in buildings including converting waste into something useful, effecting the development of CEBOs from WP3. These workshops will be accompanied with press releases and short videos to capture media coverage and spread the learnings and doings from HOUSEFUL.

Stakeholders relations, networking and clustering:
- Participation in fairs and conferences - HOUSEFUL partners will participate in conferences and fairs;
- Results from developed solutions will be disseminated in 4 technical workshops, covering advances in buildingcircularity on energy (M30) in Barcelona (Spain), waste (M33) in Terrassa (Spain), material (M36) in Netherlands and water (M50) in Vienna (Austria).
- Clustering activities with projects and initiatives on circular economy will be fostered through the organisation of jointdissemination activities, such as, organisation of common events, workshops, production of dedicated editorials;
- A final event (in the format of a workshop, conference, or roundtable) will be organised at EU level by LEITAT, inthe framework of another planned circular economy event, where the results and services developed and demonstrated during the project will be shared with other European stakeholders.

Task 7.5 Public communication (Leader: ICONS, LEITAT; Participants: all) [M1-M54]
This task aims at producing and distributing original HOUSEFUL contents through key and targeted messages to raise awareness, acceptance and impact towards wider audiences. Public contents will be widely distributed through EU and global information multipliers and online media (such as, Alpha Galileo, Physorg, Cordis Wire, youris.com etc.) and other information platforms and thematic portals, as well as through the partners' communication tools and will include: • At least 8 independent articles and interviews around project-related topics, written by professional journalists will be produced and distributed to information multipliers, online newspapers, dedicated portals and promoted via social media networks.
- HOUSEFUL videos. The audiovisual strategy of the project will include: one Video News Release (VNRs) forTV distribution and four web videos. Focusing on innovative outcomes related to the Front-runners buildings and raising media and public interest the VNR will be produced and distributed to European and Word TV stations through youris.com audiovisual platform, managed by ICONS, and by exploiting the satellite gateways of the Eurovision department of the European Broadcasting Union. One short web video (30 seconds – 1 minute) including animations, infographics and real footage will be produced on the HOUSEFUL demonstration outcomes.
- Press and News Releases: at least 10 press and news releases focusing on specific project issues and milestones,promoting project events and progress, will be produced and distributed. They will be made available to local partners for translation into local languages and distribution through their existing channels.
- Pageflow – a digital storytelling tool, enabling to incorporate many editorial formats (background videos, images,text…) into a richer narrative illustrating and transforming the HOUSEFUL outcomes into market-able solutions.

Task 7.6. HOUSEFUL impact monitoring and assessment. (Leader: ICONS) [M1-M54]
Measuring outreach, community engagement and gathering citizens and stakeholders' feedback is essential to assess the effectiveness of the designed D&C strategy and HOUSEFUL impacts. A continuous monitoring activity of the effects of the widespread distribution of the HOUSEFUL contents on web and social media channels via analytics and dedicated state-of-the-art software tools or through direct data retrieval via the information multipliers' platforms

will enable the project to measure its outreach. Outreach measurement is aimed at the definition of the absolute impact data, which provide an input to the Community Engagement Index analysis. The Community Engagement Index (CEI) measures the actual engagement of the project community with the project contents delivered on the internet and on social media. The CEI creates a univo-cal relation between any project content made available on the web and the actual interactions made by visitors coming across that content. The weighted aggregation of all CEI values calculated on single project con-tents will then generate the Project Engagement Index (PEI) representing the engagement capacity of the entire project communication and a fundamental key to the definition of the impacts of the project.

| Participation per Partner | |
|---|---|
| **Partner number and short name** | **WP7 effort** |
| 1 - LEITAT | |
| 2 - ITeC | |
| 3 - CARTIF | |
| 4 - ALCN | |
| 5 - AHC | |
| 6 - VISUM | |
| 7 - AIGUASOL | |
| 8 - LGI | |
| 9 - GWR | |
| 10 - Turntoo B.V. | |
| 11 - Neues Leben | |
| 12 - HOUSING EUROPE | |
| 13 - WE&B | |
| 14 - ICONS | |
| 15 - IDP | |
| 16 - HOMEBIOGAS LTD | |
| 17 - IRIS | |
| **Total** | |

List of deliverables

| Deliverable Number[14] | Deliverable Title | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|
| D7.1 | Dissemination & Communication plan | 14 - ICONS | Report | Confidential, only for members of the consortium (including the Commission Services) | 43 |
| D7.2 | HOUSEFUL website | 14 - ICONS | Websites, patents filling, etc. | Public | 5 |

List of deliverables

| Deliverable Number[14] | Deliverable Title | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|
| D7.3 | Project presentation video | 14 - ICONS | Websites, patents filling, etc. | Public | 8 |
| D7.4 | Call-To-Action video | 14 - ICONS | Websites, patents filling, etc. | Public | 18 |
| D7.5 | Flyers (Communication materials for stake-holder engagement) | 14 - ICONS | Report | Public | 8 |
| D7.6 | Project leaflet | 14 - ICONS | Report | Public | 10 |
| D7.7 | Report on public communication, networking and clustering activities (version I) | 14 - ICONS | Report | Public | 18 |
| D7.8 | Multimedia tools: 1 PageFlow, 1 VNR and 1 web video produced and distributed | 14 - ICONS | Websites, patents filling, etc. | Public | 50 |
| D7.9 | Best Practices book | 14 - ICONS | Websites, patents filling, etc. | Public | 52 |
| D7.10 | Communication and dissemination activities impact assessment | 14 - ICONS | Report | Confidential, only for members of the consortium (including the Commission Services) | 54 |

| | | | | | |
|---|---|---|---|---|---|
| D7.11 | Report on public communication, networking and clustering activities (version II) | 14 - ICONS | Report | Public | 36 |
| D7.12 | Report on public communication, networking and clustering activities (version III - final) | 14 - ICONS | Report | Public | 54 |

| Description of deliverables |
|---|
| Deliverables: D7.1) Dissemination & Communication plan [M4 with internal updates at M18, M30, M43, ICONS] / D7.2) HOUSEFUL website [M5. ICONS] / D7.3) Project presentation video [M8, ICONS] / D7.4) Call-To-Action video (Communication materials for stakeholder engagement) [M18, ICONS] / D7.5) Flyers (Communication materials for stakeholder engagement) [M8, ICONS] / D7.6) Project leaflet [M10, ICONS] / D7.7) Report on public communication, networking and clustering activities [M18, ICONS] / D7.8) Multimedia tools: 1 PageFlow, 1VNR and 1 web video produced and distributed [M50. ICONS] / D7.9) Best Practice book [M52, ICONS] / D7.10) Communication and dissemination activities impact assessment [M54, ICONS] / D7.11) Updated version of D7.7 [M36, ICONS] / D7.12) Updated and final version of D7.11 [M54, ICONS] |

D7.1 : Dissemination & Communication plan [43]

Definition and monitoring of a common and well organized communication strategy, identifying target audiences (closely related to stakeholders' identification and analysis in task 3.1), defining key messages to be conveyed per target group and establishing the most important communication formats, actions and channels, in accordance with article 28 of the GA ("Promoting the action-visibility of EU Funding"). The plan will be updated periodically throughout the project. This deliverable is related to Task 7.1.

D7.2 : HOUSEFUL website [5]

Design and launch of the HOUSEFUL website, where a description of the project (objectives and expected impacts), news, progress and milestones accomplished, direct connection with the SaaS solution developed in WP6 and other related external platforms and initiatives will be available for consultation. This deliverable is related to Task 7.3.

D7.3 : Project presentation video [8]

A short animation video addressing multiple stakeholders with key basic messages, facts and features of the HOUSEFUL project and its foreseen impacts in the circular economy of buildings will be produced to present the project in an easy-to-understand way. The video will be accessible on the website, promoted on social media and used in events to introduce the project. This deliverable is related to Task 7.3.

D7.4 : Call-To-Action video [18]

A specific video format based on animations and infographics aimed at engaging stakeholders in frontrunner buildings and inviting them to undertake actions will be produced to support citizen and stakeholder engagement activities. This deliverable is related to Task 7.2.

D7.5 : Flyers (Communication materials for stake-holder engagement) [8]

The flyer will inform local citizens and stakeholders about the HOUSEFUL solutions demonstrated in the project, including the SaaS, with simple and easy-to-understand messages. The flyer will have a common visual identity, but contents will be customized according to the specificity of each Frontrunner building. This deliverable is related to Task 7.2.

D7.6 : Project leaflet [10]

The HOUSEFUL leaflet, addressing European stakeholders at large, will provide information on the main characteristics of the project and its expected results and impacts. Key key information and messages will be adapted to properly engage the different stakeholders identified in Task 3.1. This deliverable is related to Task 7.2.

D7.7 : Report on public communication, networking and clustering activities (version I) [18]

Report regarding the activities related to communication strategies deployed during the project with respect to public communication, including articles and interviews, videos (D7.3, D7.4 and D7.9), press and news releases and pageflow (D7.8); networking and clustering activities, such as participation in fairs and conferences, activities with other circular economy projects and events attended and organized by the project. This deliverable is related to Task 7.4.

D7.8 : Multimedia tools: 1 PageFlow, 1 VNR and 1 web video produced and distributed [50]

Production of a PageFlow (digital storytelling tool, enabling to incorporate many editorial formats into a richer narrative to enhance HOUSEFUL marketable solutions), a Video News Release to disseminate the project's main outcomes addressing the broad audience via TV media and a web video summarizing the main project outcomes. This deliverable is related to Task 7.5.

D7.9 : Best Practices book [52]

The Best Practices book will provide an overview of the main project achievements, best practices, lessons learnt and recommendations. Contents will be retrieved from the main outcomes of the demonstration activities (described in the deliverables of the different WPs) and will be packaged in a direct and effective graphic layout. The Best Practices book will be available in both electronic and printable format. This deliverable is related to Task 7.4.

D7.10 : Communication and dissemination activities impact assessment [54]

Report on the impacts achieved through the implementation of the Dissemination and Communication Plan and Strategy established at the beginning of the project (D7.1). Continuous monitoring activity of outreach and community engagement (both online - through dedicated indicators - and off-line through qualitative analysis) will provide measures to assess the project impacts in terms of awareness, engagement and acceptance. This deliverable is related to Task 7.6.

D7.11 : Report on public communication, networking and clustering activities (version II) [36]

| |
|---|
| Updated version of D7.7<br>D7.12 : Report on public communication, networking and clustering activities (version III - final) [54] Updated and final version of D7.11 (being D7.11 and update of D7.7) |

**Schedule of relevant Milestones**

| Milestone number[18] | Milestone title | Lead beneficiary | Due Date (in months) | Means of verification |
|---|---|---|---|---|
| MS12 | Definition of the Dissemination and Communication Strategy | 14 - ICONS | 4 | D7.1) The Dissemination and Communication Plan is finalised and approved at consortium level |
| MS13 | Project website online | 14 - ICONS | 5 | D7.2) The HOUSEFUL website is available in internet |

| Work package number [9] | WP8 | Lead beneficiary [10] | 1 - LEITAT |
|---|---|---|---|
| Work package title | Ethics requirements | | |
| Start month | 1 | End month | 54 |

## Objectives

The objective is to ensure compliance with the 'ethics requirements' set out in this work package.

## Description of work and role of partners

**WP8 - Ethics requirements** [Months: 1-54]
**LEITAT**
This work package sets out the 'ethics requirements' that the project must comply with.

## List of deliverables

| Deliverable Number[14] | Deliverable Title | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|
| D8.1 | H - Requirement No. 1 | 1 - LEITAT | Ethics | Confidential, only for members of the consortium (including the Commission Services) | 1 |
| D8.2 | POPD - H - Requirement No. 3 | 1 - LEITAT | Ethics | Confidential, only for members of the consortium (including the Commission Services) | 1 |
| D8.3 | GEN - Requirement No. 4 | 1 - LEITAT | Ethics | Confidential, only for members of the consortium (including the Commission Services) | 3 |

## Description of deliverables

The 'ethics requirements' that the project must comply with are included as deliverables in this work package.

D8.1 : H - Requirement No. 1 [1]

The procedures and criteria that will be used to identify/recruit research participants must be submitted as a deliverable by the end of M1, before the start of activities involving humans. The applicant must clarify whether vulnerable individuals/groups will be involved, and the measures to protect them and minimise the risk of their stigmatisation must be submitted as a deliverable by the end of M1, before the start of activities involving humans.

D8.2 : POPD - H - Requirement No. 3 [1]

The informed consent procedures that will be implemented for the participation of humans and for the collection, storage, and protection of personal data, must be submitted as a deliverable by the end of M1, before the start of activities involving humans or collection of personal data. Templates of the informed consent/assent forms and information sheets (in language and terms intelligible to the participants) must be submitted as a deliverable by the end of M1, before the start of activities involving humans or the collection of personal data.

D8.3 : GEN - Requirement No. 4 [3]
An independent Ethics Advisor must be appointed to monitor the ethics issues involved in this project and how they are handled. The Advisor must be consulted at least on the following points: data handling, vulnerable people, consent and recruitment. A report by the Ethics Advisor must be submitted as preliminary scoping document at the end of month 6 to indicate the procedures and ethical requirements in the project. There must then be a final report to be submitted at M54. It is recommended that the ethics advisor is involved in the project throughout.

## Schedule of relevant Milestones

| Milestone number[18] | Milestone title | Lead beneficiary | Due Date (in months) | Means of verification |
|---|---|---|---|---|

### *1.3.4. WT4 List of milestones*

| Milestone number[18] | Milestone title | WP number[9] | Lead beneficiary | Due Date (in months)[17] | Means of verification |
|---|---|---|---|---|---|
| MS1 | Methodology for evaluating the circularity level of buildings | WP2 | 7 - AIGUASOL | 26 | D2.5) The methodology defined in WP2 will be validated using real scenarios (all Frontrunner buildings) and integrated in the SaaS developed in WP6. |
| MS2 | Desk research of existing circular solutions & services | WP2 | 7 - AIGUASOL | 26 | D2.4) Existing market circular solutions |
| MS3 | Social engagement plan | WP3 | 13 - WE&B | 26 | Stakeholder analysed process finished for all buildings, related to D3.1) Social engagement strategy. |
| MS4 | Co-created services | WP3 | 13 - WE&B | 48 | Engagement workshop organised, related to D3.3. |
| MS5 | Baseline of Frontrunner buildings completed | WP4 | 3 - CARTIF | 23 | D4.1) D4.11) Baseline calculation for 4 Frontrunner buildings completed |
| MS6 | Execution project documents ready | WP4 | 3 - CARTIF | 36 | D4.9) Technical specifications and execution projects of Frontrunner buildings |
| MS7 | Technological solutions ready to be validated at demonstration scale | WP4 | 3 - CARTIF | 36 | D4.4) Optimisation of technological solutions (Task 4.3), Financial scheme (task 4.6), and technical monitoring programme (task 4.8). |
| MS8 | Environmental, economic and social assessment of the circular solutions demonstrated | WP5 | 1 - LEITAT | 54 | D5.2) Environmental and economic impact assessment; D5.3) Measurement &Verification; D5.4) Social assessment |
| MS9 | Launch of the market study | WP6 | 8 - LGI | 24 | D6.1) Market study |
| MS10 | Launch of the SaaS | WP6 | 8 - LGI | 36 | D6.3) SaaS ready to be used by stakeholders |
| MS11 | Complete list of potential HOUSEFUL services for stakeholders/end-users | WP6 | 8 - LGI | 54 | D6.7) Final Exploitation Plan |
| MS12 | Definition of the Dissemination and Communi-cation Strategy | WP7 | 14 - ICONS | 4 | D7.1) The Dissemination and Communication Plan is finalised and approved at consortium level |

| | | | | | |
|---|---|---|---|---|---|
| MS13 | Project website online | WP7 | 14 - ICONS | 5 | D7.2) The HOUSEFUL website is available in internet |

| Milestone number[18] | Milestone title | WP number[9] | Lead beneficiary | Due Date (in months)[17] | Means of verification |
|---|---|---|---|---|---|
| MS14 | Financial plan for Demo 2 | WP4 | 3 - CARTIF | 26 | In task 4.6, delivery of D4.5 due on M26 and updated version on M48 (D4.8). It includes a Financing plan, a progress monitoring of the implementation of the plan, risk/uncertainty assessment and mitigation plans. |
| MS15 | Financial Close for Demo 3 | WP4 | 3 - CARTIF | 30 | Complementary financial access for demo 3 is completed. Associated deliverables - D4.6 (in M30) and updated version D4.8 (M48). |
| MS16 | Tenants occupy the "Els Mestres" building | WP4 | 3 - CARTIF | 40 | They begin to participate in co-creation activities (D3.5) |

### *1.3.5. WT5 Critical Implementation risks and mitigation actions*

| Risk number | Description of risk | WP Number | Proposed risk-mitigation measures |
|---|---|---|---|
| 1 | Difficult in identifying the key parameters for quantifying the circularity level of residential buildings | WP2, WP6 | A robust and reliable methodology to quantify the circularity level will be developed based on calculation methods and indicators which take into account a life-cycle approach. Partners' expertise will assure the successful definition and quantification of KPIs. |
| 2 | A robust and reliable methodology to quantify the circularity level is not feasible to be developed based on calculation methods and indicators which take into account a life-cycle approach. | WP3 | A robust and reliable methodology to quantify the circularity level will be developed based on calculation methods and indicators which take into account a life-cycle approach. Partners' expertise will assure the successful definition and quantification of KPIs. The parameters should be measurable and easy to quantify from data available from current draws and data about building characteristics. The change of the Demo 2 and Demo 3 should not impact on this Risk. On the other hand, if punctual data for quantification is missing, collaboration with dwellers will be essential. In this case, the new Demo 2 and Demo 3 offer different and positive willingness to collaborate: - Demo 2 ("Sant Quirze" building): Part of the neighbourhood is already collaborating in tasks related with Plug-n-harvest project (Block A). Current impact and willingness for collaboration is positive. Community living in other blocks potentially used for renovating/ improving deficiencies (Bloc B1 and B2) using HOUSEFUL can be opened to innovation actions considering the previous experience of neighbours in Plug-n-harvest project. Co-creation for data acquisition (if required) seems to be feasible. -<br>Demo 3 ("Cambium" building): The dwellers are concerned about ecology, sustainable development and circularity, which they have a lot knowledge and therefore interested and willing to collaborate. Co-creation for data acquisition (if required) seems to be feasible. |
| 3 | Low participation of stakeholders in co-creation workshops | WP3 | A very thorough engagement strategy, framework and protocol will be developed at the start of the project to ensure a high level of engagement of the project stakeholders. If participation is nevertheless low, the engagement plan will be revised. Co-creation with dwellers will be essential for the successful development and demonstration of services. The identification of needs and beliefs requires a first contact with stakeholders and the neighbourhood. Depending on the willingness to participate in HOUSEFUL, co-creation will be a success or not. As previously mentioned for Demo 2 ("Sant Quirze" building), there is a positive experience of part of the (block<br>A) neighbours participating in Plug-n-harvest |

| Risk number | Description of risk | WP Number | Proposed risk-mitigation measures |
|---|---|---|---|
| | | | project (GA:768735). A similar or a higher motivational collaboration in co-creation activities is expected by co-creating with other (Block B1 and B2) neighbours. Moreover, a positive impact is expected for this building. HOUSEFUL co-creation methodology can be integrated in the execution Plug-n-harvest project. Then a synergistic effect between both projects can be generated. For Demo 3 ("CAMBIUM" building) communication with dwellers is very open for cocreation, as they are all engaged in the topic of a sustainable community. |
| 4 | Delays in demonstration activities | WP4 | All actions committed in HOUSEFUL are aligned to the action plans of the building owners/tenants. Long term plans supported by already committed funds and a solid ecosystem of private partners that are strongly committed to the project are the measures to mitigate the impact of this risk. When necessary, funding opportunities will be searched and applied (WP4-task 4.6). Regarding task 4.6, periodic review of the progress monitoring of the imple-mentation of the financing plan will be done, including a risk/uncertainty assessment and realisation of mitigation plans. All actions committed in HOUSEFUL are aligned to the action plans of the building owners/tenants. Long term plans supported by already committed funds and a solid ecosystem of private partners that are strongly committed to the project are the measures to mitigate the impact of this risk. When necessary, funding opportunities will be searched and applied (WP4-task 4.6). Regarding task 4.6, periodic review of the progress monitoring of the implementation of the financing plan will be done, including a risk/uncertainty assessment and realisation of mitigation plans. The change of Demo 2 and Demo 3 impacts on this risk: - Demo 2 ("San Quirze" building): The confirmation of this building as the new demo-site shall involve the preparation of the tender process near the initial schedule. On the other hand, renovation works for "Sant Quirze" considering HOUSEFUL approach and social participation can be performed faster than expected in "La Gripia" building. Main reason is the different type of work to be performed in facades, which needs less budget and working time than the one expected for roof renovation (in "La Gripia"). Then, the overall impact of this risk for this building is positive. - Demo 3 ("Cambium" building): Demonstration of S1-S3 and S6-S8 are possible, with preparation of the detailed planning activities in M26, if agreed on. Lack of budget and plans for overall refurbishment could |

| | | | |
|---|---|---|---|
| | | | result in postponed demonstration of S4 and S9-S10 |

| Risk number | Description of risk | WP Number | Proposed risk-mitigation measures |
|---|---|---|---|
| | | | but could either be feasible towards the end of the project duration or we will deliver a plan and roadmap for optimum implementation afterwards. |

| | | | |
|---|---|---|---|
| 5 | Planned investments not materialised in demo 1 and 2 | WP4 | Investment plans foreseen, in which innovative business models accompanying each specific action were included, demonstrate the strong commitment of both public bodies and private companies to implement the activities in three two Frontrunner buildings. Search of new funding sources, review of the financing plan, progress monitoring, risk/ uncertainty assessment and review of mitigation plans. AHC has increased the planned investment for renovation of Demo 1 by using a total 2.2M€ own funding (+0.7M € than expected), making possible the tender process (published in January 2019). Because the tender process for Demo 1 has been launched, this risk (number 5) does not affect the Demo 1. The change of the Demo 2 impacts on this Risk. AHC had considered an initial budget and commitment of 1.2M€ for renovating the asbestos roof of "La Gripia" building. Considering that a recent technical diagnosis of "La Gripia" roof discarded this renovation need, the economic impact for AHC has been significantly reduced. The limited monetary resources of AHC restricts the availability of funding in renovation projects. These projects are executed in order of priority and determined according to the social needs and the building characteristics: AHC must refurbish the B1 and B2 blocks of "San Quirze" building with 24 homes, due to the important pathologies (in façade, damps and deficiencies in the supply of hot water supported by solar panels). A total of 260.400€ is the cost/budget of the planned renovation, to be covered by own funding. If this building is accepted as Demo 2 in HOUSEFUL, renovation will be performed using HOUSEFUL principles. |
| 6 | Implementation of NBS does not fit into vegetation period | WP4 | Pre-grow plants in nurseries, adjustment of timetable The change of the Demo 2 and Demo 3 does not impact on this risk. The solution S6 related to treatment of un-segregated water using NBS+disinfection for reuse is planned to be demonstrated in Demo 2 and Demo 3. The strategy to follow with the implementation and use of plants should be the same for both buildings. |
| 7 | Low biogas production or quality | WP4 | Review of lab tests regarding C/N/P ratios according to new blackwater:BW mixture. Testing of alternative inoculums. Review of biogas cleaning system for proper dimensioning of |

| Risk number | Description of risk | WP Number | Proposed risk-mitigation measures |
|---|---|---|---|
| | | | energy conversion system. The change of the Demo 2 and Demo 3 does not impact on this Risk. |

| | | | |
|---|---|---|---|
| 8 | Digestate with low nutrient content | WP4 | Use of digestate as raw material for cocomposting using gardening waste or kitchen wastes. |
| 9 | Low water reuse potential by high evapotranspiration | WP4 | Review of NBS design. Selection of plants adapted to regional climate conditions where NBS are implemented. Better control of inlet water flow in NBS. |
| 10 | Lack of environmental data to assess the environmental impact of implemented solutions | WP5 | In case of not obtaining enough environmental data from BIM, scientific literature and existing LCA databases such as Ecoinvent or GaBi Professional will be used. The report related to this task is the assessment plan (reported in M17). Once Demo 2 and Demo 3 are selected as new building alternatives, the plan would be executed accordingly. The related results will be reported at the following reporting period and at the end of the project execution. The delay in the development of previous tasks and deliverables (i.e. D4.1) would not influence on the WP5 execution. |
| 11 | Not sufficient inputs for the market assessment | WP6 | Identify with all project partners a broad list of stakeholders to be consulted. The change of the Demo 2 and Demo 3 slightly impacts on this Risk. Stakeholders related to the Demo 2 and Demo 3 will be identified and consulted together under the framework of WP3 and Task 3.1 [M1-M26]. |
| 12 | Low participation of stakeholders in knowledge sharing activities (i.e. technical workshops, training sessions, study tours) | WP7 | A very thorough communication strategy will be developed at the start of the project to ensure a high level of visibility of the project for the targeted audience and will seek for a strong involvement of the stakeholders. If participation is nevertheless low, the communication plan will be revised. The change of the Demo 2 and Demo 3 slightly impacts on this Risk. Contingency will be activated if low participation of new stakeholders (technical stakeholder, not dwellers) occurs. Specially for Demo 3, we would also need to activate additional stakeholders reflecting the local communities. We believe, that uptake of our planned technologies might be easier in more rural areas, as people are still more connected to nature's cycles and will be more willing to reintegrate human nutrients for their food production. |
| 13 | Technical failure in the demo-sites | WP4 | Several demo-sites have been prepared in order to guarantee an efficient and smooth process of demonstration. The change of the Demo 2 and Demo 3 impacts on this risk. The demonstration of appropriate solutions will depend on the |

| Risk number | Description of risk | WP Number | Proposed risk-mitigation measures |
|---|---|---|---|
| | | | alternative building selected as well as the cocreation activity developed with dwellers. |

| | | | |
|---|---|---|---|
| 14 | Demonstrated technological solutions (water-S5,S6; waste- S7,S8,S9) includes high investment and/or O&M cost | WP4 | Review of technical specifications, designs and dimensioning of each technology at pilot scale. Evaluation of alternative circular materials for construction. Reviewing O&M costs. Search of additional funding schemes for improved feasibility. The change of the Demo 2 and Demo 3 will not impact on this Risk. Low cost and circular materials for an efficient O&M cost of technological solutions (water-S5, S6; waste- S7, S8, S9) will be sought whatever the final building selected. |
| 15 | AnMBR installation in buildings is not permitted due to biogas risk issues | WP4 | A risk assessment will be conducted during unit design in order to define whether ATEX equipment is required in order to fulfil installation requirements. Additional CHP unit will be considered to avoid biogas storage, directly transforming it to electricity. |
| 16 | Low interest, lack of understanding from the stakeholders' perception (not acceptance) and misuse of new technologies and the SaaS. | WP3, WP6, WP7 | The project will adopt a scalable approach to the dissemination and engagement strategy. Thanks to continuous monitoring process of dissemination, communication and social acceptance indicators it will be possible to implement corrective and customised measures (new dissemination tools, 1-to-1 communications, focussed meetings) to ease understanding and knowledge sharing and increase the level of interest among stakeholders. The project will adopt a scalable approach to the dissemination and engagement strategy. Thanks to continuous monitoring process of dissemination, communication and social acceptance indicators it will be possible to implement corrective and customised measures (new dissemination tools, 1-to-1 communications, focused meetings) to ease understanding and knowledge sharing and increase the level of interest among stakeholders. The change of the Demo 2 and Demo 3 impacts on this Risk. In detail for Demo 2, the fact that "Sant Quirze" building and related stakeholders are familiarized with European projects by PLUG-AN HARVEST project (GA: 768735), the performance of co-creation activities can be easier than expected for LA GRIPIA building but with a major impact at European scale. On the other hand for the Demo 3, due to the social and ecological character of the community living in "Cambium" building, the performance of co-creation activities can be expected to be positive. |
| 17 | The project is ambitious and it could undermine the credibility | WP1, WP2, WP3, WP4, WP5, WP6, WP7 | The extensive expertise of all partners of HOUSEFUL guarantees a high chance of success the achievement of project objectives. Moreover, a strong project management progress interaction |

| **Risk number** | **Description of risk** | **WP Number** | **Proposed risk-mitigation measures** |
|---|---|---|---|

| | | | |
|---|---|---|---|
| | | | (thorough three-monthly updates or throughout extra technical meetings or conference calls) will ensure the suitable progress of the project and thus the feasibility of HOUSEFUL. If any technical delay or problem is occurred during the project, scientific coordination along with advisory board will give the most proper solution for achieving the minor dispersion of the objectives. |
| 18 | Planned investment not materialised for Demo site 3 | WP4 | Several other co-financing schemes will be sought during the project execution including Vienna city funding, funding for being part of international fairs like the International Building Exhibition which will take place in 2022 in Vienna, other H2020 projects, further nationally funded projects with the Austrian Research Promotion Agency, bank loans with European guarantees, etc. If no additional funding would be available, most of the technology demonstration will be incorporated in other ways. For Demo site 3, only the innovative winter garden will be difficult to implement without deep refurbishment. All the other technologies in Demo 3, including biogas, green walls, green house, composting system, incorporation of nutrients, etc., will, in the worst case scenario, be realised in the same scale at the same site without full refurbishment of the building. We will still separate the wastewater and use all the organic solid residues of the whole building for biogas generation and further composting and all liquid residues in green infrastructures and nutrient use in urban farming. The incorporation of rainwater is planned mostly within the innovative winter garden, but if this cannot be installed the rainwater can only be used for conventional garden irrigation. In this case we would lose one innovation, which can also be realised in a smaller scale as double window embedment in about 0,5m2 surface area instead of originally planned 12m2. We can still use this window system as smart window and regulation of indoor air ventilation. It will not have the same cleaning effects as there will be space for three plants instead of 40 plants in the large setup. This innovation is currently being developed in the running project ReCO2ST (Residential Retrofit assessment platform and demonstrations for near zero energy and CO2 emissions with optimum cost, health, comfort and environmental quality. H2020-EeB-05-2017, grant agreement nr. 768576) and will be demonstrated in Denmark and the UK. For “Cambium” building, we so far plan to increase the size of the water/nutrient treatment and reuse setups as the building and the community is bigger and they could benefit more from a demonstration site, that better |

| Risk number | Description of risk | WP Number | Proposed risk-mitigation measures |
|---|---|---|---|
| | | | suits their size. We will be able to implement bigger NBS installations including the green walls, the biogas and composting systems and potentially a wintergarden, housing the NBS and farming area. On the other hand, it is unclear when the wholesome refurbishment of the building will start. Since the Cambium building is located in Styria, local funding opportunities (subsidies) have to be exploited. Fehring is part of the Vulkanland, a network of municipalities, to support the future viability of the region, which also focuses on sustainable solutions in the housing sector, water management and agriculture. There is also an opportunity to apply for additional 'asset' pool, with new investors. We will anyways support the selected alternative to our best possibilities and will try to guide them towards a circular refurbishment process and will also support them to get additional funding. |

### *1.3.6. WT6 Summary of project effort in person-months*

| | WP1 | WP2 | WP3 | WP4 | WP5 | WP6 | WP7 | WP8 | Total Person/Months per Participant |
|---|---|---|---|---|---|---|---|---|---|
| 1 - LEITAT | | | | | | | | | |
| 2 - ITeC | | | | | | | | | |
| 3 - CARTIF | | | | | | | | | |
| 4 - ALCN | | | | | | | | | |
| 5 - AHC | | | | | | | | | |
| 6 - VISUM | | | | | | | | | |
| 7 - AIGUASOL | | | | | | | | | |
| 8 - LGI | | | | | | | | | |
| 9 - GWR | | | | | | | | | |
| 10 - Turntoo B.V. | | | | | | | | | |
| 11 - Neues Leben | | | | | | | | | |
| 12 - HOUSING EUROPE | | | | | | | | | |
| 13 - WE&B | | | | | | | | | |
| 14 - ICONS | | | | | | | | | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 15 - IDP | | | | | | | | | |
| 16 - HOMEBIOGAS LTD | | | | | | | | | |
| 17 - IRIS | | | | | | | | | |
| **Total Person/Months** | | | | | | | | | |